Midterm Review*

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October 21, 2026

Abstract

We study why organizations conduct interim performance reviews when monetary rewards are limited. An interim review creates incentive capacity by allowing future work to reward past performance. Optimal policies map continuous performance scores into a discrete incentive ladder: termination, difficult or easy continuation, and, for exceptional performance, early award of the maximal prize for success—interpretable as a promotion or a grant of job security—with no further work. Review can sustain high effort when terminal compensation alone cannot. Its timing balances two forces: waiting improves performance information but leaves less future work available to motivate the agent.

Keywords: feedback, bounded reward, incentive ladder, incentive capital.

*We thank Ralph Boleslavsky, Liang Dai, Felix Feng, Canice Prendergast, Peter Norman Sørensen, Ming Yang, Yenan Wang and John Zhu for helpful comments and suggestions. All remaining errors are our own.

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1 Introduction

Performance reviews are ubiquitous in organizations and often have substantial consequences for an employee's career. Evaluations affect bonuses and merit pay, but also promotion, retention, and dismissal (Cappelli and Conyon, 2018). This raises a basic question: if an organization can ultimately observe an informative measure of performance and reward the employee accordingly, why evaluate performance before the end of the current phase of employment? In a standard moral-hazard model with unrestricted monetary transfers, there is no need to: it is optimal to bundle incentives from different stages into a sufficiently powerful terminal compensation scheme.

This paper shows that the answer changes fundamentally when monetary rewards are bounded. Let M denote the largest performance-contingent payment that the principal can make. The cap may be literal, but more broadly it captures organizations in which large financial bonuses are difficult, costly, or institutionally constrained. Such restrictions are particularly natural in public-sector employment, where pay is relatively compressed and promotion is an important source of incentives (Mas, 2017; Deserranno, Kastrau, and León-Ciliotta, 2025), and in nonprofit organizations, which make less use of output-contingent incentive pay than for-profit firms (DeVaro and Brookshire, 2007). More generally, many careers provide substantial incentives through advancement rather than immediate cash compensation. Tenure and promotion in academia and promotion to partnership in law, accounting, consulting, and other professional-service firms are prominent examples (Baker, Jensen, and Murphy, 1988; Barlevy and Neal, 2019; Levin and Tadelis, 2005). In these environments, the opportunity to continue on a favorable career path is itself a reward.

This observation gives interim review a role that is absent when monetary incentives are unbounded. A review allows the principal to allocate future opportunities according to past performance. An agent who performs well can be offered a more attractive continuation contract, while one who performs poorly can face a demanding continuation standard or termination. Rents that must be conceded to motivate future effort can do double duty: besides inducing future effort, they can be used as rewards for past effort. We refer to these future rents and continuation opportunities as *incentive capital*. When M is large, the principal can instead provide incentives directly with money and incentive capital has little value. When M is tight, continuation opportunities become valuable because monetary rewards are scarce.

We study this mechanism in a canonical dynamic moral-hazard environment. A risk-neutral agent produces cumulative Gaussian output over a finite horizon and privately chooses

costly effort.[1] A risk-neutral principal receives output net of compensation, which is subject to limited liability and bounded above by M. The principal may commit to one public interim review, choose when it occurs, and condition both compensation and the continuation opportunity on the review score. Past performance contains information about past effort but does not change the technology, productivity, or difficulty of future work. Thus review has no sorting, learning, or technological role in the baseline: any value it creates comes purely from incentives.

Our first main result is that a continuum of possible review scores is optimally mapped into at most four discrete actions. These actions form an *incentive ladder*, ordered by the rent they deliver to the agent. At the bottom is a *bad stop*: sufficiently poor performance leads to termination with no payment. The next rung is continuation under a difficult terminal standard, which induces future effort while leaving the agent relatively little rent. Better review performance earns continuation under an easier terminal standard and hence a larger rent. Finally, when first-phase incentives are sufficiently scarce, exceptional performance reaches a *good stop*: the agent receives the maximal payment M immediately and is released from the remaining workload. Thus termination can optimally occur at both tails of the performance distribution. The bad stop is the strongest punishment available; the good stop is the strongest reward available once money alone can no longer reward performance. Depending on parameters, two, three, or all four rungs are used.

The discreteness of this ladder is notable because both performance and the feasible set of continuation contracts are continuous. Its structure comes from the geometry of bounded incentive provision. Among contracts that induce future effort, the principal uses only the minimum- and maximum-rent continuation contracts. Together with the two extreme no-effort actions—termination without pay and termination with the maximal payment—these are the only extreme points of the set of feasible continuation actions. Monotone likelihood ratios then order the four actions by review performance. In this sense, a continuous performance evaluation optimally produces a coarse rating system with only a small number of economically distinct consequences.

Our second set of results establishes when review is valuable. The payment bound M is again the key comparative static. When M is sufficiently large, no genuine review can improve on the terminal contract: uniformly over all review dates, the principal strictly prefers to wait until the end. As the cap tightens, however, the terminal contract exhausts its ability to provide incentives and continuation rents become increasingly valuable. When the effort cost

[1] For ease of exposition, we assume that effort in the baseline model is chosen only twice, before the performance review and after it. We relax this assumption and allow the agent to choose effort at each instant in Section 6.1.

is moderate and the cap is near the smallest level at which full effort can be implemented without review, a sufficiently late review is strictly profitable and uses all four rungs of the incentive ladder. Indeed, review can do more than reduce the cost of incentives. It can make high effort implementable when terminal evaluation alone cannot. At the no-review implementability boundary, splitting the project into two shorter incentive problems creates strict incentive slack in each phase. For payment caps slightly below that boundary, full effort is impossible under any terminal-only contract but can be induced with an interim review. Review therefore expands the organization's effective incentive capacity: future rents that would otherwise be an unavoidable cost of motivating later effort become an additional currency with which to motivate earlier effort.

Our third main result concerns *when* the review should occur. Delay has an obvious informational advantage. With more performance accumulated before the review, the distributions generated by working and shirking are easier to distinguish. But waiting also uses up the very future opportunities that make review valuable. A later review leaves less remaining work, a smaller range of continuation rents, and therefore less incentive capital with which to reward first-phase performance. Optimal review timing balances better information against more time left to motivate.

The endpoint results make this tradeoff particularly sharp. An arbitrarily early review is strictly harmful: the right derivative of review value at date zero is $-\infty$. A very short probationary period generates almost no useful information, while bounded rewards prevent the principal from cheaply magnifying that information into incentives. Any optimal genuine review is therefore bounded away from the initial date. At the other endpoint, when M is tight and the effort cost moderate, moving the review slightly before the terminal date has first-order value because even a short continuation phase creates scarce incentive capital. Thus the model predicts neither "review as soon as possible" nor "wait until performance is most informative." The optimum is governed by the tension between learning about past effort and preserving future opportunities with which to reward it.

We also examine three departures from the baseline. First, we allow the agent to choose effort at every instant; the four-rung ladder continues to apply, although the easy continuation standard becomes more demanding. Second, we study the opposite informational extreme in which output is publicly observed continuously and the agent can react to the entire path. The optimal contract then becomes a continuous-time band contract with firing and buyout boundaries. This benchmark also shows that more frequent review need not benefit the principal: over a substantial range of payment caps, committing to one optimally timed review dominates continuous observation. Finally, we show that random review timing can be valuable because it relaxes the sharp tradeoff between collecting more pre-review information

and preserving post-review incentive capital.

Related Literature This paper contributes to the literature on interim performance evaluation and dynamic incentive provision by asking three questions: Why review? How should review information be used? And when to review?

First, we identify a new rationale for interim review. As in Lizzeri, Meyer, and Persico (2002), we consider a setting in which past performance does not change the primitive continuation agency problem. With unrestricted monetary rewards, the principal can bundle incentives across the two phases into a single terminal contract, making review unnecessary. However, when payments are bounded, as in the static setting of Jewitt, Kadan, and Swinkels (2008), review allows the principal to condition otherwise unavoidable continuation rents and the remaining workload on earlier performance, thereby creating history dependence and expanding incentive capacity. In contrast, much of the literature considers environments in which earlier actions or outcomes affect the difficulty or return of subsequent work. In such settings, an important role of interim review is to reveal the severity of the continuation agency problem and allow the continuation contract to respond accordingly (see, e.g., Lukas, 2010; Chen and Chiu, 2013; Ely and Szydlowski, 2020). A separate rationale is efficient sorting: Ray (2007) uses interim evaluation to identify and terminate projects with poor continuation prospects. Relatedly, Liu (2026) studies resource allocation when the agent privately knows a project's fixed quality and always prefers a larger allocation. A public midterm signal correlated with project quality allows the principal to discipline the agent's initial report, effectively "levering" information from the review into the first-period allocation. Our setting has neither private information about project quality to elicit nor heterogeneous continuation prospects to uncover. Instead, review provides information about the agent's hidden productive effort, and its value comes from using continuation decisions to provide incentives when monetary rewards are limited.

Second, we show that interim performance can be used through a simple, cutoff-based contract. Depending on interim performance, the agent is terminated or retained under a continuation contract that pays only following sufficiently good final performance. Termination can therefore occur at both tails: a bad stop punishes poor performance, while a good stop rewards strong performance by combining the maximal monetary payment with relief from future effort. Whereas Ely, Georgiadis, and Rayo (2025) jointly design dynamic incentives and the disclosure of performance feedback, our cutoff structure concerns how revealed interim performance is translated into termination and continuation incentives.

Good-performance termination also arises in other dynamic moral hazard settings (see, e.g., Spear and Wang (2005) and Sannikov (2008)), but for a different reason. In those models,

favorable performance raises the promised value of a risk-averse agent until maintaining incentives becomes too costly, so termination occurs when continuation value reaches a sufficiently high level. Our good stop is instead a direct response to the payment cap. Once the maximal monetary prize has been reached, reducing the agent's remaining workload provides the additional reward that money cannot. This logic is closer to Zhu (2013), where periods of shirking can optimally serve as relaxation periods that reward good performance.

Third, we identify a new tradeoff governing the timing of review. Delaying review has a familiar benefit: more accumulated performance makes the review signal more informative. But delay has a novel cost in our setting: it shortens the remaining production horizon and thereby reduces the future output that can be used to create continuation rents. Review timing therefore determines not only the quality of information available to the principal, but also the amount of incentive capacity remaining after the review.

This question connects our paper to a recent literature on the design and timing of dynamic monitoring. Varas, Marinovic, and Skrzypacz (2020) show that optimal monitoring can take the form of deterministic periodic reviews when information acquisition is important, but random inspections when monitoring primarily serves to provide incentives. Similarly, Ball and Knoepfle (2026) study whether inspections should be predictable or random in a long-term moral-hazard relationship, with the answer depending on whether effort primarily promotes breakthroughs or prevents breakdowns. Wong (2026) goes further by allowing the principal to design the monitoring technology itself, obtaining a nonstationary sequence of pass/fail tests whose informativeness and consequences evolve over the relationship. Relatedly, Dai, Wang, and Yang (2026) jointly design monitoring and compensation, with the principal dynamically reallocating limited monitoring capacity between evidence confirming effort and evidence revealing shirking. These papers endogenize the pattern, composition, or technology of monitoring, whereas we take a single interim review as given and ask where it should be placed within a finite production horizon.

Closest to our timing question, Rodivilov (2022) studies when to monitor an agent engaged in experimentation and shows that monitoring has both a static effect on contemporaneous moral hazard and a dynamic effect on the rents that must be provided in earlier periods. Our mechanism is different. In our model, the benefit of delay comes from the increasing informativeness of accumulated performance, while its cost comes from exhausting the very continuation opportunities through which review provides incentives.

This distinction also separates our mechanism from Hoffmann, Inderst, and Opp (2021) and Georgiadis and Szentes (2020), who study the timing of performance measurement following a one-time hidden action rather than an interim review during ongoing production. In Hoffmann, Inderst, and Opp (2021), delaying payout improves the information on which

compensation is based but entails a direct cost of deferral; in Georgiadis and Szentes (2020), continued monitoring improves precision but incurs a direct monitoring cost. In our model, by contrast, the cost of delay arises from the contracting technology itself: waiting for a more informative review leaves less future work with which to reward favorable performance and provide incentives.

2 Environment

A risk-neutral principal (she) contracts with a risk-neutral agent (he) over a finite horizon $[0, T]$. Output X_t evolves as $dX_t = e_t\, dt + dB_t$, with $X_0 = 0$, where B is a standard Brownian motion and $e_t \in \{0, 1\}$ is agent effort. We write $\Phi(\cdot)$ for the standard normal distribution and $\phi(\cdot)$ for its density.[2]

The contract fixes at date 0 a single review date $\tau \in (0, T]$.[3] The review date is ultimately a choice variable, and we solve for it in two steps: Section 4 takes τ as *exogenous*, and Section 5 then *endogenizes* it.

Assumption 1 (Observability of the output)**.** *Cumulative output is publicly observed at τ and at T. No one, including the agent, observes X_t at any other time.*

The review splits the horizon into two phases, $P_1 = (0, \tau]$ and $P_2 = (\tau, T]$, of lengths τ and $T - \tau$; we call X_τ the *review score*. The case $\tau = T$ is the *no-review benchmark*: the second phase is empty and X_T is the only signal.

Assumption 2 (Phase-level private effort)**.** *Effort is constant within each phase. The agent privately chooses $a_1 \in \{0, 1\}$ at date 0 and, having observed X_τ, privately chooses $a_2 \in \{0, 1\}$ at date τ; effort is $e_t = a_1$ on P_1 and $e_t = a_2$ on P_2. A strategy is a pair (a_1, S) with $S : \mathbb{R} \to \{0, 1\}$ giving $a_2 = S(X_\tau)$.*[4]

Under Assumption 2, the two phase increments are independent given (a_1, a_2), with

$$X_\tau \sim N(a_1\tau, \tau), \qquad X_T - X_\tau \sim N\Big(a_2(T - \tau),\, T - \tau\Big).$$

Working through a phase of length h costs the agent ch, so total effort cost is $c\Big[a_1\tau + a_2(T - \tau)\Big]$ with $c \in (0, 1)$. After observing X_τ and X_T the principal makes a measurable payment $w(X_\tau, X_T)$ to the agent.

[2] Gaussianity is not essential for the main results. A general binary statistical experiment with suitable two-sided-evidence and monotone-likelihood-ratio conditions yields qualitatively similar findings. Details available upon request.

[3] The limiting case $\tau = 0$ is used only as a boundary limit in Section 5.

[4] Assumption 2 is relaxed in Section 6.1, where the agent chooses $e_t \in \{0, 1\}$ at each instant, so that a phase of length h delivers any total effort $\eta = \int e_t\, dt \in [0, h]$ rather than only $\{0, h\}$.

Assumption 3 (Limited liability and a payment cap)**.** *The payment function satisfies* $w : \mathbb{R}^2 \to [0, M]$ *with* $M \in (0, \infty)$.

The lower bound of 0 is the agent's limited liability, and the upper bound M is a finite cap: the maximum payment the principal can make. The principal receives ex post payoff $X_T - w$, and the agent receives $w - c\Big[a_1\tau + a_2(T - \tau)\Big]$. At $t = 0$ the principal chooses a contract that consists of a review date τ and a payment function w to maximize her expected payoff. Under the no-review benchmark $\tau = T$ a contract is simply $w(X_T) \in [0, M]$. The principal has all the bargaining power and full commitment, and the agent's outside option is zero.

3 Contracting over a single phase

We begin the analysis with the single-phase contract. Fix a phase of length $h > 0$ at the end of which output is observed and payment is made, with no further review. Two cases of the model have this form, $h = T$ (no-review) and $h = T - \tau$ (post-review); in the latter, translation invariance of the technology lets a continuation contract be written as a function of the second-phase output $Y = X_T - X_\tau$. Solving the phase problem once therefore delivers both cases.

Under work $Y \sim N(h, h)$ and under shirking $Y \sim N(0, h)$, with densities $f_j^h(y) = h^{-1/2}\phi\Big((y - jh)/\sqrt{h}\Big)$ for $j \in \{0, 1\}$; working costs the agent ch. A contract is a measurable $w : \mathbb{R} \to [0, M]$; it implements work if

$$\mathbb{E}_1[w] - \mathbb{E}_0[w] \;\geq\; ch, \tag{IC$_h$}$$

where $\mathbb{E}_j[w] = \int w f_j^h$. Let $\mathcal{F}_h$ denote the set of contracts satisfying (IC$_h$). Throughout the section c and h are fixed, and we suppress the dependence of derived objects on (h, c, M) except where an argument varies.

3.1 Threshold contracts

First, we show that it is without loss to focus on *threshold contracts*: pay the cap M when output clears a bar b and nothing below it. The reason is that the likelihood ratio $\ell^h(y) = f_0^h(y)/f_1^h(y) = \exp(h/2 - y)$ is strictly decreasing, so a contract that shifts payment toward high output — rewarding evidence of work and withholding payment after evidence of shirking — provides incentives at the lowest expected cost.

Lemma 1. *Let $w : \mathbb{R} \to [0, M]$ be any contract and let $\tilde{w} = \mathbf{1}_{\{y \geq b\}} M$ be the threshold contract with the same expected cost under work, $\mathbb{E}_1[\tilde{w}] = \mathbb{E}_1[w]$. Then $\mathbb{E}_0[\tilde{w}] \leq \mathbb{E}_0[w]$, with equality only if $w = \tilde{w}$ almost everywhere. Hence $w \in \mathcal{F}_h$ implies $\tilde{w} \in \mathcal{F}_h$.*

Nothing is therefore lost by restricting attention to threshold contracts. Moreover, because every threshold contract pays either 0 or M, it can equivalently be interpreted as awarding a fixed prize M (e.g., a promotion) whenever performance clears the bar.

Now, index the bar b by the *normalized* value $z = (h - b)/\sqrt{h}$, so that a larger z is an easier bar, cleared with probability $\Phi(z)$ under work. For $w = \mathbf{1}_{\{y \geq b\}} M$,

$$\mathbb{E}_1[w] = M\Phi(z), \qquad \mathbb{E}_0[w] = M\Phi\Big(z - \sqrt{h}\Big), \qquad \mathbb{E}_1[w] - \mathbb{E}_0[w] = M\,\Delta(h, z), \tag{1}$$

where $\Delta(h, z) = \Phi(z) - \Phi(z - \sqrt{h})$ is the *incentive index*: the increase in the probability of clearing the bar when the agent works rather than shirks. By (1) a threshold contract implements work exactly when

$$\Delta(h, z) \;\geq\; \frac{ch}{M}. \tag{2}$$

Figure 1 illustrates the admissible bars. The index is single-peaked at $z = \sqrt{h}/2$, where it attains its maximum $\overline{\Delta}(h) = 2\Phi\Big(\sqrt{h}/2\Big) - 1$, symmetric about that peak, and vanishes as $z \to \pm\infty$. Work is therefore implementable if and only if $M \;\geq\; M_I(h, c) \equiv \frac{ch}{\overline{\Delta}(h)}$. When it holds, the admissible bars form the closed interval $[z_D, z_E]$ with

$$z_D = \min\left\{z : \Delta(h, z) \geq \frac{ch}{M}\right\}, \qquad z_E = \max\left\{z : \Delta(h, z) \geq \frac{ch}{M}\right\}, \tag{3}$$

the two crossings of the horizontal line at height ch/M, which merge at $\sqrt{h}/2$ when $M = M_I(h, c)$. The constraint binds at z_D and z_E and is slack strictly between them. The lower crossing z_D corresponds to a high-output bar and sets a *difficult* standard b_D; the upper crossing z_E corresponds to a low bar and sets an *easy* one b_E.

Implementability does not guarantee that the principal wants the agent to work: the rent she must concede to do so may exceed the surplus the phase generates. A second threshold therefore governs profitability.

Lemma 2. *There is a threshold $\underline{M}(h, c) \geq M_I(h, c)$ such that work is implementable and the principal's payoff from implementing it is nonnegative if and only if $M \geq \underline{M}(h, c)$. Additionally, there exists $\bar{c}(h) \in (0, 1)$ such that $\underline{M}(h, c) = M_I(h, c)$ for all $c \leq \bar{c}(h)$; otherwise $\underline{M}(h, c) > M_I(h, c)$.*

Appendix A characterizes $\bar{c}(h)$ and $\underline{M}(h, c)$ explicitly. We impose the following assumption to ensure that contracting without a review is strictly profitable.

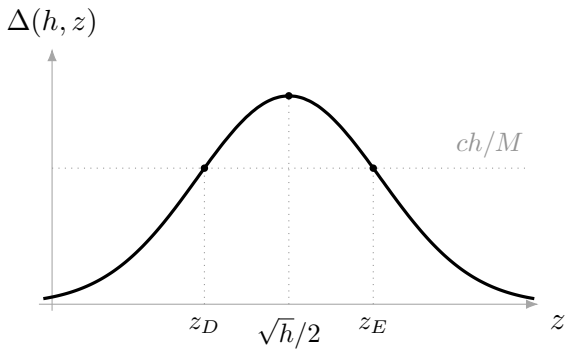


Figure 1: The incentive index $\Delta(h,\cdot)$

Assumption 4 (Implementable and profitable full effort)**.** $M > \underline{M}(T,c)$.

3.2 The no-review benchmark

Setting $h = T$ gives the benchmark against which a review must be evaluated: the principal writes a single contract on terminal output and never intervenes. By Lemma 1 she may take it to be a threshold contract, and among the admissible bars she chooses the cheapest, which is the difficult one.

Proposition 1. *Under Assumption 4, the principal's optimal no-review payoff is* $V^{\mathrm{NR}}(T;c,M) = T - M\Phi\Big(z_D(T;c,M)\Big) > 0$*, attained by the difficult bar. Moreover* $z_D(T;c,M) \to -\infty$ *and* $V^{\mathrm{NR}}(T;c,M) \to (1-c)T$ *as* $M \to \infty$*, so the first-best surplus accrues to the principal as the payment cap is relaxed.*

The limit is what makes a review worth studying. Since $z_D = (T - b_D)/\sqrt{T}$, z_D falling to $-\infty$ corresponds to a bar b_D rising to $+\infty$ in output: as M grows the principal promises an ever larger prize on an ever less likely outcome. Because the Gaussian likelihood ratio is unbounded, that trade is favorable for the principal. The agent's rent is what he could secure by shirking, $\mathbb{E}_0[w] = M\Phi\Big(z_D - \sqrt{T}\Big)$ by (1), and as the bar rises this vanishes even though M diverges: the reward event becomes overwhelming evidence of work. A principal who can promise arbitrarily large payments therefore captures the whole surplus without ever consulting interim performance, and has nothing to gain from a review. When M is finite the agent retains a strictly positive rent, and the question is whether the principal can do better by making the continuation depend on an interim signal.

3.3 The review action set

Now take $h = T - \tau$ to be a continuation phase. After a review score x the principal's choice is a continuation contract $w(x, \cdot) : \mathbb{R} \to [0, M]$ on the second-phase output Y, and the agent's best response to it determines whether second-phase work occurs. For everything that follows, such a contract is summarized by two numbers: the expected continuation utility r it leaves the agent net of effort cost, and the *total* expected continuation surplus s it generates—the principal's own expected continuation payoff is then $u = s - r$. We call (r, s) an *action*, describe the review by the set of available actions, and drop the qualifier 'expected' when no confusion arises. When effort is implemented, total surplus is $(1 - c)h$, so the only remaining question is which rents the agent can be promised.

Lemma 3. *Suppose $M > M_I(h, c)$. The rents attainable while implementing work are exactly $[r_D, r_E]$, where*

$$r_j = M\Phi(z_j) - ch = M\Phi\Big(z_j - \sqrt{h}\Big), \quad j \in \{D, E\}, \qquad 0 < r_D < r_E < M - ch,$$

and $r_D + r_E = M - ch$.

Here r_D and r_E are the rents left by the difficult and the easy standard respectively. The principal's payoff from the difficult standard, $(1 - c)h - r_D$, equals $V^{\mathrm{NR}}(T; c, M)$ when $h = T$. The set of rent–surplus pairs available under work is therefore $\mathcal{A}_1 = [r_D, r_E] \times \{(1 - c)h\}$.

If work is not implemented, surplus is zero, and since feasible payments range over $[0, M]$ the corresponding set is $\mathcal{A}_0 = [0, M] \times \{0\}$. The feasible action set is therefore $\mathcal{A} = \mathcal{A}_0 \cup \mathcal{A}_1$. A review menu assigns one such action to every review score. Thus, after each score, the principal chooses both a continuation profile—no work from $\mathcal{A}_0$ or work from $\mathcal{A}_1$—and the rent delivered within that profile. The next section determines how these choices should vary with the review score in order to implement first-phase effort.

4 The exogenous review date

Having established what can be offered after the review, we now move backward to the incentives that precede it. Throughout this section Assumption 4 holds, the review date $\tau \in (0, T)$ is fixed exogenously, and we write $h = T - \tau$.

4.1 The optimal incentive ladder

Let g_1^τ and g_0^τ be the densities of the review score $X_\tau \sim N(a_1\tau, \tau)$ under first-phase work ($a_1 = 1$) and shirking ($a_1 = 0$). Their likelihood ratio $g_1^\tau(x)/g_0^\tau(x) = \exp(x - \tau/2)$ is strictly

increasing.

Let $a(x) = \big(r(x), s(x)\big) \in \mathcal{A}$ be the action assigned after review score x, delivering the agent rent $r(x)$ and the principal payoff $s(x) - r(x)$. Working through the first phase yields the agent $\int r\, g_1^\tau - c\tau$ and shirking yields $\int r\, g_0^\tau$, so pre-review obedience requires

$$\int_{\mathbb{R}} r(x)\,[g_1^\tau(x) - g_0^\tau(x)]\,dx \geq c\tau, \tag{OB$_\tau$}$$

which also guarantees participation, since $r \geq 0$ makes the utility from shirking nonnegative. Conditional on inducing first-phase work, expected first-phase output is τ, so the principal's problem at $t = 0$ is

$$\begin{aligned} V^R(\tau) = \sup_{a(\cdot)} \quad & \tau + \int_{\mathbb{R}} \big[s(x) - r(x)\big] g_1^\tau(x)\,dx \\ \text{subject to} \quad & \text{(OB}_\tau\text{)}, \quad a(\cdot) \text{ measurable}, \quad a(x) \in \mathcal{A} \quad \text{for a.e. } x. \end{aligned} \tag{P$_\tau$}$$

Proposition 2. *Fix $\tau \in (0, T)$ and suppose first-phase work is implemented. An optimal menu takes the ordered form*

$$a^*(x) = \begin{cases} A_B, & x < k_0^*, \\ A_D, & k_0^* \leq x < k_1^*, \\ A_E, & k_1^* \leq x < k_2^*, \\ A_G, & x \geq k_2^*, \end{cases}$$

where $A_B = (0, 0)$, $A_D = \big(r_D, (1-c)h\big)$, $A_E = \big(r_E, (1-c)h\big)$, and $A_G = (M, 0)$ are the four extreme points of $\mathrm{co}(\mathcal{A})$, and $k_0^* \leq k_1^* \leq k_2^* \leq +\infty$, with strict inequality between any two finite cutoffs.

Proposition 2 is our first main result. It shows that the optimal contract partitions the review scores into at most four ordered regions, assigning continuation profiles with progressively higher rents as performance improves. After the lowest performance realizations, A_B supplies the strongest punishment: the relationship ends with no payment and no second-phase production.[5] At intermediate performance realizations the principal preserves the continuation surplus $(1-c)h$. She first uses the difficult standard A_D, which delivers the smallest rent consistent with continuation effort, and then the easy standard A_E, which raises the agent's rent from r_D to r_E without sacrificing production. Only after the highest

[5] Because both parties are risk-neutral, the outcome $A_B = (0, 0)$ can be implemented by either a hard stop in which the principal terminates the agent and X_t ceases to evolve, or a soft stop in which the agent shirks out the clock.

performance realizations can the principal move to A_G, paying the maximal rent M but giving up the continuation surplus altogether. This progression linking better review outcomes with higher discrete rent levels is what we refer to as the *incentive ladder*.

The optimal ladder therefore uses continuation contracts as intermediate incentive instruments. Before resorting to the maximal monetary reward, it increases the agent's continuation rent while keeping second-phase effort in place. The weak ordering of the cutoffs allows some of the upper profiles (rungs of the ladder) to be absent.

4.2 Deriving the optimal ladder

To understand why the optimal menu has this form, return to Problem (P_τ). This is an optimization over the entire menu $a(\cdot)$, and its only constraint linking choices across review scores is pre-review obedience. Despite its infinite-dimensional appearance, the problem admits an elementary geometric solution. This approach also reveals the economic forces governing the choice among stopping and continuation arrangements.

Let $\lambda^* \geq 0$ be a Lagrange multiplier on (OB_τ) at the optimum and define the *net shadow value of continuation rent*

$$Q_\lambda(x) \equiv \lambda \left(1 - \frac{g_0^\tau(x)}{g_1^\tau(x)}\right) - 1 = \lambda \left[1 - \exp\left(\frac{\tau}{2} - x\right)\right] - 1. \tag{4}$$

Adjoining (OB_τ) and factoring out $g_1^\tau(x)$ gives the Lagrangian

$$\mathcal{L}(a; \lambda, \tau) = \tau - \lambda c\tau + \int_{\mathbb{R}} g_1^\tau(x) \Big\{ s(x) + Q_\lambda(x)\, r(x) \Big\} dx. \tag{5}$$

For each λ, the Lagrangian is maximized over measurable menus with $a(x) \in \mathcal{A}$ for almost every x, and the optimal menu maximizes (5) at $\lambda = \lambda^*$, together with primal feasibility and complementary slackness. This reformulation separates the two agency problems: post-review incentive compatibility and payment feasibility are encoded in $\mathcal{A}$, while the multiplier brings pre-review obedience into the objective.[6] Because the Lagrangian is additively separable across review scores, choosing a menu reduces to maximizing $s + Q_\lambda(x)r$ over $\mathcal{A}$ for almost every x, where $Q_\lambda(x)$ is the weight on the agent's continuation rent after score x. For $\lambda > 0$, $Q_\lambda(x)$ increases with x, as the work-to-shirking likelihood ratio does, so better performance tilts the choice toward higher-rent continuation profiles. By Lemma B.1(iii) in Appendix B, $\lambda^* > 0$ outside a knife-edge case, so this characterization applies at the optimum.

Although the original action choice at a given review score is from $\mathcal{A} = \mathcal{A}_0 \cup \mathcal{A}_1$, it is

[6] This is the familiar likelihood-ratio formulation of a static moral-hazard problem (Holmström, 1979), except that the reward is a continuation profile rather than a wage.

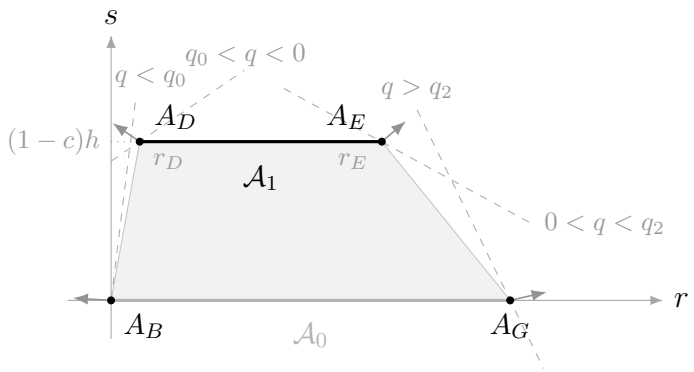


Figure 2: The review action set and the supporting lines of the assignment objective

convenient to solve the linear program

$$a_\lambda(x) \in \arg\max_{(r,s)\in \mathrm{co}(\mathcal{A})} \Big\{ s + Q_\lambda(x) r \Big\}.$$

Because the objective is linear in (r, s), convexification does not change the maximal value.

A linear objective over the compact polygon $\mathrm{co}(\mathcal{A})$ is maximized at an extreme point, and its four extreme points A_B, A_D, A_E and A_G all belong to $\mathcal{A}$. Hence a maximizer can always be chosen from $\mathcal{A}$, and public randomization cannot improve the principal's payoff; at a breakpoint an entire edge may be optimal, but so are its endpoints. Figure 2 depicts this problem: the bold segments form $\mathcal{A}$, the shaded trapezoid is $\mathrm{co}(\mathcal{A})$, and the dashed level sets of $s + qr$ are moved in the direction $(q, 1)$ until they support the hull.

Fix $q = Q_\lambda(x)$. The four vertices deliver

$$v_B(q) = 0, \qquad v_D(q) = (1-c)h + q\, r_D, \qquad v_E(q) = (1-c)h + q\, r_E, \qquad v_G(q) = q\, M,$$

whose slopes $0 < r_D < r_E < M$ are ordered, so the upper envelope moves toward higher-rent actions as q rises. Adjacent vertices tie at

$$q_0 = -\frac{(1-c)h}{r_D} \;<\; q_1 = 0 \;<\; q_2 = \frac{(1-c)h}{M - r_E}, \tag{6}$$

where the level sets of $s + qr$, of slope $-q$, are parallel to the edges $A_B A_D$, $A_D A_E$ and $A_E A_G$ of Figure 2; concavity of the upper boundary of $\mathrm{co}(\mathcal{A})$ gives the ordering. The maximizing action is therefore A_B for $q < q_0$, A_D for $q_0 < q < q_1$, A_E for $q_1 < q < q_2$, and A_G for $q > q_2$. Since $\lambda^* > 0$, $Q_{\lambda^*}(x)$ is strictly increasing in x, so as performance rises q passes through these intervals in order, yielding the ordered regions of Proposition 2.

4.3 On the form of the optimal ladder

We now ask which rungs the optimal ladder uses: when termination follows poor or strong performance, and when continuation is split into difficult and easy standards.

Corollary 1. *Suppose* $(1-c)h \neq r_D$.[7] *Whenever first-phase work is implemented,* k_0^* *is finite and the bad-stop and difficult-standard regions are nonempty. Moreover:*

[(i)]

(i) If $1 + q_0 < \lambda^* \leq 1$, *then* $k_1^* = k_2^* = +\infty$, *and only the bad-stop and difficult-standard regions are nonempty.*

(ii) If $1 < \lambda^* \leq 1 + q_2$, *then* $k_0^* < k_1^* < +\infty$ *and* $k_2^* = +\infty$, *so the bad-stop, difficult-standard, and easy-standard regions are nonempty.*

(iii) If $\lambda^* > 1 + q_2$, *then* $k_0^* < k_1^* < k_2^* < +\infty$, *and all four regions are nonempty.*

The multiplier λ^* is the shadow value of the first-phase obedience constraint. It measures the principal's marginal gain from relaxing the incentive requirement for first-phase work. A large multiplier therefore means that this requirement is particularly costly to satisfy: first-phase incentives are scarce, and the principal is willing to use stronger rewards following favorable performance.

This interpretation is reflected directly in the menu. The index $Q_{\lambda^*}(x)$ sweeps the interval $(-\infty, \lambda^* - 1)$ as performance x ranges over $\mathbb{R}$, and it crosses breakpoint q_j if and only if $\lambda^* > 1 + q_j$. Implementing first-phase work requires $\lambda^* > 1 + q_0$, so the bad stop and the difficult standard are always used. Once $\lambda^* > 1$, the incentive value of additional rent is high enough to justify the easy standard, which rewards stronger performance while preserving second-phase production. Once $\lambda^* > 1 + q_2$, even the easy standard provides too little rent at the top: the principal then uses the good stop, paying the maximal reward despite sacrificing the continuation surplus. Whenever a breakpoint is reached, the corresponding cutoff is the unique solution to $Q_{\lambda^*}(k_j^*) = q_j$.

An especially sharp implication emerges when the review occurs late.

Proposition 3. *For every review date* τ *sufficiently close to* T *from below,* $k_2^*(\tau) < +\infty$.

For any finite admissible cap, a sufficiently late review uses all four rungs. To see the force behind this result, recall that the good stop becomes active when $\lambda^* > 1 + q_2$. As the continuation horizon $h = T - \tau$ vanishes, the surplus sacrificed by replacing the easy standard

[7] The knife-edge case $(1 - c)h = r_D$, in which a zero supporting multiplier may arise, is discussed in Appendix B.

with the good stop shrinks faster than the additional rent thereby created. Hence $q_2(h) \to 0$. At the same time, the fixed-review shadow value converges to the no-review shadow value, which is strictly greater than one under Assumption 4. Thus the good-stop threshold is eventually crossed, and all four regions are active.

5 The timing and value of review

In this section, the principal chooses the review date endogenously from $\tau \in (0, T]$. A date $\tau \in (0, T)$ represents a genuine interim review. A review at $\tau = 0$ is equivalent to one at $\tau = T$, so we use $\tau = T$ to represent no review and do not consider $\tau = 0$ separately. We first study how the value of a genuine review varies with its date and then determine when such a review dominates no review.

5.1 The optimal genuine review date

For $h = T - \tau > 0$ and shadow value q, define the continuation support function by

$$\mathcal{V}(q, h) \equiv \max\{0,\ (1-c)h + q\, r_D,\ (1-c)h + q\, r_E,\ qM\}. \tag{7}$$

Complementary slackness implies that the optimized value equals the Lagrangian at the constrained optimum. Hence

$$V^R(\tau) = \tau - \lambda^* c\tau + \int_{\mathbb{R}} g_1^\tau(x)\mathcal{V}\Big(Q_{\lambda^*}(x), T - \tau\Big)\, dx,$$

where $Q_{\lambda^*}(x)$ is the net shadow value of continuation rent defined in Equation (4).

Lemma 4. *Suppose Assumption 4 holds. Then*

$$\lim_{\tau \downarrow 0} V^R(\tau) = \lim_{\tau \uparrow T} V^R(\tau) = V^{\mathrm{NR}}(T; c, M).$$

Both boundaries degenerate, for opposite reasons. As $\tau \downarrow 0$ the review score carries almost no information about first-phase effort, so there is nothing to condition on; as $\tau \uparrow T$ the continuation phase vanishes, so there is no continuation opportunity left to allocate. Either way the principal is left with the terminal contract. The no-review benchmark is therefore the right normalization across the whole interval, and any gain from review must be an interior phenomenon.

The tradeoff among genuine review dates. At points of differentiability (away from knife-edge parameters), the envelope theorem, holding the multiplier and the selected actions fixed, decomposes the derivative as

$$\frac{dV^R(\tau)}{d\tau} = \underbrace{1-\lambda^* c}_{\text{first-phase production}} \underbrace{- \int_{\mathbb{R}} g_1^\tau(x) \frac{\partial \mathcal{V}}{\partial h}\Big(Q_{\lambda^*}(x), h\Big) dx}_{\text{incentive capital}} + \underbrace{\int_{\mathbb{R}} \left[\frac{\partial g_1^\tau(x)}{\partial \tau} \mathcal{V}\Big(Q_{\lambda^*}(x), h\Big) + g_1^\tau(x) \frac{\partial \mathcal{V}}{\partial q}\Big(Q_{\lambda^*}(x), h\Big) \frac{\partial Q_{\lambda^*}(x)}{\partial \tau} \right] dx}_{\text{information}} . \tag{8}$$

The first term captures first-phase production. Delaying the review lengthens the first phase, adding a unit of production while tightening first-phase obedience, whose shadow value is charged at rate c. The second term is the opposing loss of incentive capital. The continuation phase is a stock of *incentive capital*: a family of work-inducing contracts that produce second-phase output while delivering different rents, so that rents conceded for second-phase incentives can be recycled into rewards for first-phase effort. As h falls, the surplus these contracts preserve vanishes, and strong first-phase rewards increasingly require sacrificing it, culminating in A_G. The third term is the information effect. It values a sharper performance measure: the review-score means under working and shirking differ by τ against a standard deviation $\sqrt{\tau}$, so their standardized separation grows with the review date; since $\partial \mathcal{V}/\partial q$ is the rent the menu assigns, precision is worth most at scores that already carry a large rent.

An interior optimum balances these three terms, setting (8) equal to zero. The optimal review date therefore does not merely trade an early noisy signal against a late precise one; it balances the precision of the review against the productive contracting capacity that survives it.

The suboptimality of extreme review dates. The next result distinguishes the implications of the review-date tradeoff near the two boundaries.

Proposition 4. *Review dates sufficiently close to the initial date are strictly suboptimal. Review dates sufficiently close to the terminal date are also strictly suboptimal when the payment cap is sufficiently tight. Formally,*

[(i)]

(i) for every τ sufficiently close to the initial date, there exists another genuine review date τ' such that $V^R(\tau') > V^R(\tau)$;

(ii) suppose $c \leq \bar{c}(T)$, so that $\underline{M}(T,c) = M_I(T,c)$. If M is sufficiently close to $\underline{M}(T,c)$

from above, then every review date τ sufficiently close to the terminal date is strictly dominated by an earlier genuine review date $\tau' < \tau$.

The first conclusion reflects the information term in (8). A short first phase adds only a small amount of production and generates an extremely weak signal of effort. To motivate first-phase effort, the principal must nevertheless create enough dispersion in continuation rents across review outcomes. Because those rents are bounded, extracting the required incentives from an almost uninformative signal is disproportionately costly. In particular, the principal must fire the agent with positive probability based on vanishingly little information. The first-phase production gain therefore cannot compensate for the incentive loss generated by a review sufficiently close to the initial date.

The second conclusion reflects the opposing incentive-capital term. Near the terminal date, delaying the review yields a more informative score but leaves little continuation surplus to preserve while rents vary. A tight payment cap makes this loss especially costly: the terminal contract is already close to exhausting its incentive capacity, so continuation rents have a high shadow value. The marginal loss of incentive capital then dominates the production and information benefits of further delay, and an earlier review raises the principal's payoff.

As a result, under the tight-cap conditions of Proposition 4, if an optimal genuine review exists, it cannot lie near either the beginning or the end of the project. It is genuinely a *midterm* review. Short probationary periods observed in practice must therefore reflect motives absent from the model, such as screening an unknown type, learning match quality, or preserving an option to abandon a bad project.

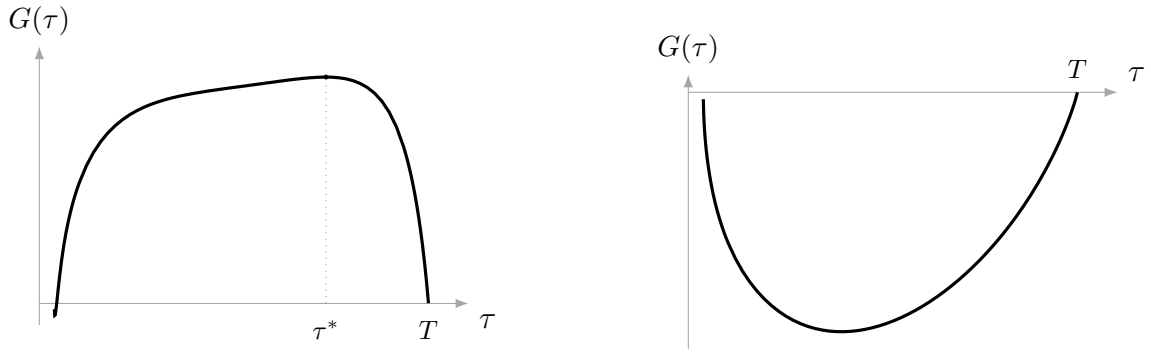
5.2 When is a genuine review valuable?

We now compare genuine review with the option of no review. Define the review gain for $\tau \in (0, T)$ by $G(\tau) = V^R(\tau) - V^{\mathrm{NR}}(T; c, M)$.

Loose payment caps. When the payment cap is loose, the no-review contract concentrates compensation on an increasingly rare terminal outcome and implements effort with vanishing rent (Proposition 1). A review cannot match this rent extraction: it either splits the terminal signal into two shorter, less precise experiments or bases incentives on an almost uninformative early signal.

Proposition 5. *Fix $T > 0$ and $c \in (0, 1)$. There exists a finite $\overline{M}(T, c)$ such that $G(\tau) < 0$ for every $\tau \in (0, T)$ whenever $M > \overline{M}(T, c)$.*

The threshold $\overline{M}(T, c)$ does not depend on the review date. A bound for each fixed τ would leave open the possibility that, as M rises, the principal keeps moving the review date



(a) Tight cap: $M = 2$.

(b) Loose cap: $M = 5$.

Figure 3: The review gain $G(\tau)$ under a tight and a loose payment cap

and stays ahead of the benchmark; Proposition 5 rules this out. No split of the horizon into a review phase and a continuation phase preserves both the precision and the production horizon of the terminal contract, so the rent compression of Proposition 1 eventually dominates at every review date.

Tight payment caps. The comparison can reverse when the payment cap is sufficiently close to the implementability boundary. The next result records the resulting local value improvement.

Proposition 6. *Suppose $c \leq \bar{c}(T)$, so that $\underline{M}(T, c) = M_I(T, c)$. If M is sufficiently close to $\underline{M}(T, c)$ from above, there exists $\delta > 0$ such that*

$$V^R(\tau) > V^{\mathrm{NR}}(T; c, M) \qquad \text{for every } \tau \in (T - \delta, T).$$

By Lemma 4, $G(\tau) \to 0$ as $\tau \uparrow T$, and under the stated tight-cap conditions the left derivative of V^R at T is strictly negative (Lemma C.1); hence $G(\tau) > 0$ just below T. This negative slope is also what drives Proposition 4(ii).

By Proposition 3, these profitable late reviews use all four rungs of the incentive ladder.

Payment caps and the value of review. Figure 3 illustrates the two regimes for $T = 1.5$ and $c = 0.6$. Under the tight cap $M = 2$, just above the implementability floor $M_I(T, c) = 1.96$, the gain reaches an interior maximum at $\tau^* = 1.089$, where $V^R = 0.239$ against $V^{\mathrm{NR}} = 0.195$; the optimal menu uses all four rungs. Under the loose cap $M = 5$, the gain is negative at every review date and the principal waits until T.

Conflict over review. There is a simple conflict of interest over the use of review. Let $U^R(\tau)$ denote the agent's equilibrium payoff under the optimal contract with review at τ, and let U^{NR} denote his payoff under the optimal no-review contract. Since no review induces work throughout the horizon, it generates total surplus $(1-c)T$. A review cannot increase total surplus and may reduce it by terminating production after some review outcomes. Hence

$$V^R(\tau) + U^R(\tau) \leq (1-c)T = V^{\mathrm{NR}}(T; c, M) + U^{\mathrm{NR}}.$$

It follows immediately that whenever review strictly benefits the principal, $V^R(\tau) > V^{\mathrm{NR}}(T; c, M)$, it strictly harms the agent: $U^R(\tau) < U^{\mathrm{NR}}$. Thus any review that the principal would voluntarily introduce is opposed by the agent relative to no review.

Review restores implementability under tight caps. Under tight caps, the value of a midterm review also lies in restoring implementability. At the no-review implementability boundary, splitting the horizon creates strict incentive slack in both phases, and a review permits full effort for caps slightly below the level required without one.[8]

Proposition 7. *Let $M_0 \equiv M_I(T, c)$, at which full effort without review is just implementable. For every split $\tau \in (0, T)$, with $h = T - \tau$,*

$$M_0\overline{\Delta}(\tau) > c\tau, \qquad M_0\overline{\Delta}(h) > ch.$$

Moreover there exist $\tau_0 \in (0, T)$ and $\varepsilon > 0$ such that for every $M \in (M_0 - \varepsilon, M_0)$ full effort is infeasible without a review, but a review at τ_0 admits a two-region menu inducing first-phase work and, whenever the project is continued, full continuation effort. Continuation occurs with positive probability.

A review can therefore create incentives unavailable when performance is evaluated only at the terminal date. Splitting the project gives two shorter incentive problems, each strictly easier to implement, since $M_0\overline{\Delta}(\tau) > c\tau$ and $M_0\overline{\Delta}(h) > ch$. More importantly, it creates a new source of motivation: the continuation phase has a nondegenerate range of work-inducing rents $[r_D, r_E]$, and the principal can condition those rents on the review score. Rents that must in any event be conceded to motivate future effort thus double as rewards for past effort, supplemented when necessary by the two stopping actions.

[8] This result steps outside Assumption 4, which requires $M > \underline{M}(T, c) \geq M_I(T, c)$. The ladder construction of Section 4 still applies at τ_0, because both phases remain strictly implementable for M close to M_0.

6 Extensions

6.1 Pointwise effort choice

The baseline model restricts the agent to a phase-by-phase effort choice: in a phase of length h, he either works throughout or does not work at all. This section allows him to choose effort at each instant. Because no new information arrives within a phase, only total effort matters. Writing $e_t \in \{0, 1\}$ for instantaneous effort, let $\eta = \int_0^h e_t \, dt \in [0, h]$. Total effort η produces $Y \sim N(\eta, h)$ and costs $c\eta$. Thus the binary choice $\{0, h\}$ in the baseline becomes the interval $[0, h]$: the agent can now "coast" by working for only part of the phase.

This extension has two sharp implications. First, the four-rung structure of the optimal review menu survives: no interior effort level becomes an extreme point of the continuation action set. Second, and more importantly, the easy continuation standard must become harder. Pointwise effort leaves the difficult continuation unchanged, but reduces the largest rent that the principal can deliver through the easy continuation. The extension therefore preserves the form of the incentive ladder while weakening one of its rungs.

From a global to a marginal incentive test. For a terminal wage w, let $\pi(\eta) = \mathbb{E}_\eta[w(Y)]$ denote expected compensation when total effort is η. A contract implements the target η exactly when no other effort level is more attractive:

$$\pi(\eta) - c\eta \;\geq\; \pi(\eta') - c\eta' \qquad \text{for every } \eta' \in [0, h]. \tag{IC$_h^{\text{flex}}$}$$

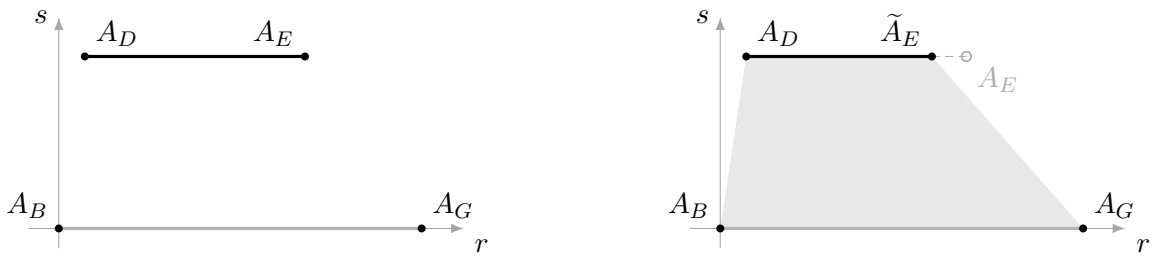
The phase-by-phase model checks only the extreme deviation $\eta' = 0$. Pointwise choice also introduces nearby ones, including the temptation to shave the last increment of effort. Two consequences of (IC$_h^{\text{flex}}$) carry the analysis: the comparison with $\eta' = 0$, and local optimality,

$$\pi(\eta) - c\eta \;\geq\; \pi(0), \tag{G$_\eta$}$$

$$\pi'(\eta) \;=\; c \;\text{ if } \eta \in (0, h), \qquad \pi'(h) \;\geq\; c \;\text{ if } \eta = h. \tag{L$_\eta$}$$

The first is the baseline's *global* work-versus-shirk condition; the second is the new *marginal* one, an equality at an interior target and an inequality at full effort, where only downward deviations are available. For the Gaussian upper-tail bonus $w(y) = M\mathbf{1}_{\{y \geq h - z\sqrt{h}\}}$ these read $\Delta(h, z) \geq ch/M$ and $\sqrt{h}\,\phi(z) \geq ch/M$, and Online Appendix OA.1 shows that the two are jointly sufficient for this threshold contract to implement full effort.

The comparison with phase-by-phase effort is now immediate. Provided the cap exceeds the (slightly higher) pointwise implementability floor of Online Appendix OA.1, so that



(a) Phase-by-phase effort, $\eta \in \{0, h\}$

(b) Pointwise effort, $\eta \in [0, h]$

Figure 4: Rent–surplus sets under phase-by-phase and pointwise effort.

$z_D(h) \leq \zeta(h)$, the global condition binds at the difficult bar $z_D(h)$ and the marginal condition is slack. The difficult continuation is therefore unchanged. At the easy bar $z_E(h)$, by contrast, the baseline global condition binds but the marginal return to the last increment of effort is too small. The reason is simple: an easy bar is cleared with high probability even if the agent works slightly less, so coasting barely changes the probability of payment while saving effort cost. A contract can therefore induce "work rather than shirk" and still fail to induce "work all the way."

The principal removes this temptation by raising the easy standard. In the z notation a larger z means an easier output bar, so the new bar is the positive root $\zeta(h)$ of

$$\sqrt{h}\,\phi\Big(\zeta(h)\Big) = \frac{ch}{M}, \qquad \text{equivalently} \qquad \phi\Big(\zeta(h)\Big) = \frac{c\sqrt{h}}{M}, \tag{9}$$

which satisfies $\zeta(h) < z_E(h)$; the associated easy action is

$$\widetilde{A}_E = \Big(\widetilde{r}_E(h), (1-c)h\Big), \qquad \widetilde{r}_E(h) = M\Phi\Big(\zeta(h)\Big) - ch \;<\; r_E(h). \tag{10}$$

Thus pointwise effort does not merely tighten full-effort feasibility; it specifically lowers the maximum continuation rent available at the easy end.

Figure 4 displays the central comparison. Moving from phase-by-phase to pointwise effort fills out the feasible rent–surplus set, but its convex hull still has four relevant vertices. Three are unchanged: bad stop A_B, difficult continuation A_D, and good stop A_G. Only the maximum-rent full-effort endpoint moves, from A_E to $\widetilde{A}_E$. Geometrically, the full-effort face is shortened; economically, the easy continuation loses incentive capital.

The four-rung menu survives. Online Appendix OA.1 proves that the convex hull of the pointwise feasible action set $\mathcal{A}^{\text{flex}}(h)$ is $\text{co}\{A_B, A_D, \widetilde{A}_E, A_G\}$: although pointwise effort permits a continuum of targets, no interior $\eta \in (0, h)$ maximizes the principal's score-by-score linear

objective. The optimal menu can therefore still be chosen from four actions, ordered by review performance exactly as in the baseline—bad stop, difficult continuation, easy continuation, good stop—and all baseline formulas carry over after the economically meaningful substitution $r_E(h) \mapsto \widetilde{r}_E(h)$. In particular the upper breakpoint becomes $\widetilde{q}_2(h) = (1-c)h/\Big(M - \widetilde{r}_E(h)\Big)$, while the breakpoints q_0 and q_1 and the ordering of the four regions are unchanged. Since $\widetilde{r}_E(h) < r_E(h)$, the easy continuation is a weaker reward: the principal uses it less and relies relatively more on the indivisible good-stop prize. Pointwise effort therefore changes the strength, but not the architecture, of the incentive ladder. This substitution away from continuation rents toward good stopping is costly: it reduces productive continuation and lowers the principal's payoff.

6.2 Continuous observability and the limit of frequent review

The baseline commits to a single public review date. A natural question is what happens when performance can instead be reviewed arbitrarily often. We study the limiting benchmark in which the entire output path is publicly observed. This changes more than the principal's information: because the agent also sees the path, effort can be conditioned on history. Thus continuous observability should be interpreted as the limit of increasingly frequent *public* reviews.

Output continues to satisfy $dX_t = e_t\,dt + dB_t$, but now $(X_s)_{s\le t}$ is public at every date and $e_t \in \{0,1\}$ is chosen pointwise. Let W_t denote the agent's promised continuation utility. Standard continuous-time incentive arguments (Sannikov, 2008) imply that, while work is recommended, promised utility must have sensitivity at least c to innovations in output. The principal therefore uses the smallest admissible sensitivity. The resulting contract has a simple band form.

Proposition 8 (Continuous-observation band contract)**.** *Suppose output is publicly observed continuously and effort is chosen pointwise. An optimal contract can be represented by an initial promised utility $W_0^* \in [0, M]$ with the following properties.*

[(i)]

(i) While $0 < W_t < M$, the agent works and promised utility evolves as

$$dW_t = c\,dX_t, \qquad \text{so that} \qquad W_t = W_0^* + cX_t.$$

(ii) The relationship ends when W_t first hits either boundary. At $W_t = 0$ the agent is terminated with zero payment; at $W_t = M$ he receives the maximal payment M and is

released from further work. If neither boundary is reached before T, he is paid W_T at the horizon.

(iii) Writing $u = W_0/c$, $m = M/c$, and letting $v(T,u)$ be the expected time before T that a unit-drift Brownian motion starting at u remains inside $(0,m)$, the principal chooses

$$u^* \in \arg\max_{u\in[0,m]} \{(1-c)v(T,u) - cu\},$$

and sets $W_0^* = cu^*$. The optimum is unique whenever it is interior.

The contract is the continuous-information analogue of the incentive ladder. The lower and upper absorbing boundaries play the roles of the bad and good stops. Between them, however, there is no longer a pair of discrete continuation rents: every instant acts as a review, so the agent's continuation utility moves continuously with performance. The slope c is the minimum sensitivity consistent with effort, and the only initial design choice is how much "cushion" W_0^* to place between the two stopping boundaries. Appendix D proves Proposition 8; Online Appendix OA.2 gives the fuller recursive derivation and numerical details.

This limiting benchmark delivers an enlightening answer to the multiple-review question. More observation is not necessarily better for the principal, because the agent's ability to react to the running score tightens incentive compatibility history by history. At the benchmark used in Figure 3, $T = 1.5$, $c = 0.6$, and $M = 2$, the optimal single review occurs at $\tau^* = 1.089$ and gives the principal 0.239, while the continuous-observation band contract gives only 0.135; no review gives 0.195. The band contract actually leaves the agent less rent than either discrete-review contract at these parameters, but it induces only about 0.854 units of expected work because the lower boundary is hit frequently. The loss is therefore not due to excessive rent but to inefficient early termination.

The comparison varies with the payment cap. Continuous observation is attractive when M is very tight because effort can be purchased in small increments and the relationship can stop as soon as promised utility reaches a boundary. Once the cap is moderate, however, a single review strictly dominates the band contract; with a sufficiently loose cap, the optimal single-review contract itself converges to no review. For $T = 1.5$ and $c = 0.6$, the band is profitable for caps above approximately 0.670 and dominates the best contract with at most one review only up to approximately $M = 1.488$. The no-review contract becomes implementable at approximately $M = 1.958$. Thus continuous observability is most valuable precisely in the region where coarse discrete evaluation has too little incentive capacity.

6.3 Random review timing with pointwise effort

The review date has so far been announced at date 0. We now ask whether uncertainty about timing can itself be useful when effort is chosen pointwise. For tractability, the principal commits to two possible dates $0 < \tau_1 < \tau_2 < T$, with probability p on the early date, but does not reveal the realization until a review occurs. Before τ_1 the agent therefore does not know which date will apply. If no review occurs at τ_1, he learns that the review will occur at τ_2 and can reoptimize his remaining effort. Thus the lottery must remain incentive compatible after the information conveyed by non-arrival.

Put $d = \tau_2 - \tau_1$, and let r_i be the rent schedule used if review occurs at τ_i. For cumulative effort e at date τ_i, define

$$U_i(e) = \mathbb{E}\Big[r_i\Big(e + \sqrt{\tau_i}Z\Big)\Big] - ce, \qquad \widehat{U}_2(e) = \max_{m \in [e, e+d]} U_2(m), \qquad Z \sim N(0,1), \tag{11}$$

where $\widehat{U}_2(e)$ is the late-branch value, given cumulative effort e at τ_1, after the agent has inferred the review date and optimally revised his effort. Let $D_1(e) = U_1(\tau_1) - U_1(e)$ and $D_2(e) = U_2(\tau_2) - \widehat{U}_2(e)$.

Proposition 9. *Suppose Assumption 4 holds and effort is chosen pointwise. Fix $0 < \tau_1 < \tau_2 < T$ and put probability p on τ_1.*

[(i)]

(i) Full pre-review effort is sequentially incentive compatible if and only if $U_2(\tau_2) = \widehat{U}_2(\tau_1)$ and

$$pD_1(e) + (1-p)D_2(e) \geq 0 \qquad \text{for every } e \in [0, \tau_1]. \tag{12}$$

(ii) If the late branch induces full effort on its own, then (12) holds if and only if $p \leq \overline{p}$, where

$$\overline{p} = \inf_{\{e\,:\,D_1(e)<0\}} \frac{D_2(e)}{D_2(e) - D_1(e)}, \tag{13}$$

with $\overline{p} = 1$ when the set is empty.

(iii) Under the hypothesis of (ii), let P_i be the principal's payoff on branch i under full effort and let $V^{D,\mathrm{pw}}$ denote the best deterministic value among contracts inducing full pointwise effort. If $P_1 > V^{D,\mathrm{pw}} > P_2$ and

$$\underline{p} \equiv \frac{V^{D,\mathrm{pw}} - P_2}{P_1 - P_2} < \overline{p}, \tag{14}$$

then the lottery with any $p \in (\underline{p}, \overline{p}]$ strictly dominates every such deterministic contract.

The attractive early branch can tolerate only so much weight before some early coasting becomes profitable; this gives the ceiling $\overline{p}$. Beating the best deterministic review requires enough weight on that branch; this gives the floor $\underline{p}$. Random timing is valuable when the incentive ceiling lies above the payoff floor.

A numerical example shows that the interval can be nonempty. Set $T = 1.5$, $c = 0.72$ and $M = 2.60$, and take $\tau_1 = 0.0949$ and $\tau_2 = 0.8166$. Using the pointwise-feasible continuation actions $A_B, A_D, \widetilde{A}_E, A_G$ from Section 6.1, the branch payoffs are $P_1 = 0.13301$ and $P_2 = 0.03199$. The late branch induces full effort on its own and $\overline{p} = 0.47293$. A relaxation of the deterministic pointwise problem gives the upper bound $V^{D,\mathrm{pw}} \leq 0.07850$, and hence $\underline{p} \leq 0.46047 < \overline{p}$. At $p = 0.465$ the lottery is sequentially incentive compatible and yields 0.07896, strictly above that upper bound and therefore above every deterministic full-effort pointwise review.

The gain is small and the example is a witness rather than an optimized lottery. Its point is qualitative: even when the agent can coast before the first possible date, learns from the non-arrival of a review, and then optimally revises his effort, uncertainty about review timing can strictly improve on every deterministic review. Appendix D proves the proposition, and Online Appendix OA.3 gives the numerical construction.

7 Conclusion

This paper studies how a principal uses a public midterm review when effort is sequential and compensation is bounded. The review does more than evaluate past performance: it allows the principal to use future production, continuation rents, and termination as incentive instruments. The optimal menu assigns these continuation opportunities through an ordered incentive ladder, while the review date balances the informativeness of accumulated performance against the incentive value of the remaining continuation phase.

The extensions also clarify the scope of the one-review institution. Pointwise effort preserves the ordered menu but weakens the easy-continuation rung. At the opposite extreme, continuous public observation produces a band contract and shows that arbitrarily frequent review need not dominate a single optimally timed review: finer feedback also gives the agent finer opportunities to condition his effort. Random review timing provides yet another margin by pooling incentive slack across possible dates. Thus, when incentives are dynamic, the principal may benefit not from observing performance as often as possible, but from choosing carefully when to look.

A Proofs for Section 3

Recall $\overline{\Delta}(h) = \max_z \Delta(h,z) = 2\Phi(\sqrt{h}/2) - 1$ and $M_I(h,c) = ch/\overline{\Delta}(h)$ from Section 3.1, and put $\bar{c}(h) = 2 - 1/\Phi(\sqrt{h}/2) \in (0,1)$.

Proof of Lemma 1. Let $b \in [-\infty, +\infty]$ solve $\mathbb{E}_1[\mathbf{1}_{\{Y \geq b\}}] = \mathbb{E}_1[w]/M$, which exists and is unique because that probability falls continuously and strictly from one to zero (with the endpoints understood as limits), and write $\varphi = w/M$, $\varphi^* = \tilde{w}/M = \mathbf{1}_{\{y \geq b\}}$. Since $\int(\varphi - \varphi^*) f_1^h = 0$, subtracting $\ell^h(b)$ times this vanishing integral gives

$$\mathbb{E}_0[w] - \mathbb{E}_0[\tilde{w}] = M \int (\varphi - \varphi^*)\big[\ell^h - \ell^h(b)\big] f_1^h.$$

For $y > b$, $\varphi^* = 1 \geq \varphi$ and $\ell^h(y) < \ell^h(b)$; for $y < b$ both inequalities reverse. The integrand is therefore nonnegative almost everywhere, so $\mathbb{E}_0[\tilde{w}] \leq \mathbb{E}_0[w]$, with equality only if $\varphi = \varphi^*$ a.e. Subtracting from $\mathbb{E}_1[\tilde{w}] = \mathbb{E}_1[w]$ gives the incentive comparison. □

Proof of Lemma 2. Implementability. By Lemma 1 the largest attainable incentive spread is $M\overline{\Delta}(h)$, so $\mathcal{F}_h \neq \emptyset$ exactly when $M \geq M_I(h,c)$, and $[z_D, z_E]$ is nondegenerate when the inequality is strict. With $a = \sqrt{h}/2$ we have $\overline{\Delta}(h)/h = \mathcal{G}(a) \equiv [2\Phi(a) - 1]/(4a^2)$ and $\mathcal{G}'(a) = -F(a)/(2a^3)$ with $F(a) = 2\Phi(a) - 1 - a\phi(a)$; since $F(0) = 0$ and $F'(a) = (1 + a^2)\phi(a) > 0$, $\mathcal{G}$ is strictly decreasing, so $M_I = c/\mathcal{G}$ is strictly increasing in h.
Profitability. By Lemma 1 and (1) the cheapest work-inducing contract is the difficult bar, so the principal's payoff is $V(M) = h - M\Phi(z_D)$. Dividing the binding condition $M[\Phi(z_D) - \Phi(z_D - \sqrt{h})] = ch$ by $\Phi(z_D)$,

$$M\Phi(z_D) = \frac{ch}{1 - q(z_D)}, \qquad q(z) = \frac{\Phi(z - \sqrt{h})}{\Phi(z)}. \tag{A.1}$$

Since ϕ/Φ is strictly decreasing, $\frac{d}{dz}\log q > 0$, so q is a strictly increasing bijection onto $(0,1)$. As M rises, ch/M falls and z_D lies on the increasing branch of $\Delta(h,\cdot)$, so z_D falls, hence by (A.1) so does $M\Phi(z_D)$: the payoff V is strictly increasing in M, with $V \uparrow (1-c)h > 0$ as $M \to \infty$ (since $\Delta(h, z_D) = ch/M \to 0$ on the increasing branch forces $z_D \to -\infty$, so $q(z_D) \to 0$) and, as $M \downarrow M_I(h,c)$ where the two bars merge at $\sqrt{h}/2$, $V \to h(1 - c/\bar{c}(h))$.
The threshold. If $c \leq \bar{c}(h)$ then $h(1 - c/\bar{c}(h))$ is nonnegative and $V > 0$ at every $M > M_I(h,c)$; set $\underline{M}(h,c) = M_I(h,c)$. If $c > \bar{c}(h)$ it is negative, so by continuity and strict monotonicity there is a unique $M_P(h,c) > M_I(h,c)$ with $V = 0$, and $\underline{M}(h,c) = M_P(h,c)$. At that cap $M\Phi(z_D) = h$ while the binding constraint gives $M\Phi(z_D - \sqrt{h}) = (1-c)h$; dividing, z_D is the root z_P of $q(z) = 1 - c$, so $M_P = h/\Phi(z_P)$. In both cases $\underline{M} \geq M_I$, with equality when $c \leq \bar{c}(h)$. □

Proof of Proposition 1. By Lemma 1 and (1) the cheapest work-inducing contract is the difficult bar, whose expected cost under work is $M\Phi(z_D(T))$; thus, $V^{\mathrm{NR}}(T; c, M) = T - M\Phi\Big(z_D(T)\Big)$. Positivity follows from Lemma 2 under Assumption 4, so implementing work beats the zero payoff from not implementing it. The limits of V^{NR} and $z_D(T) \to -\infty$ are established in the proof of Lemma 2. □

Proof of Lemma 3. Take $w \in \mathcal{F}_h$ and let $\tilde{w}$ be the cost-matched threshold contract of Lemma 1, which also lies in $\mathcal{F}_h$; its bar therefore satisfies (2), so $z \in [z_D, z_E]$ and $M\Phi(z_D) \leq \mathbb{E}_1[w] \leq M\Phi(z_E)$. Both bounds are attained, and $r = \mathbb{E}_1[w] - ch$ is affine in w on the convex set $\mathcal{F}_h$, so the attainable rents are exactly $[r_D, r_E]$, swept by mixtures of the two extreme contracts. The second equality is (1) at a binding constraint, and $r_E < M - ch$ because $\Phi(z_E) < 1$. Finally $\Delta(h, z) = \Phi(z) + \Phi(\sqrt{h} - z) - 1$ is symmetric about $\sqrt{h}/2$, so $z_E = \sqrt{h} - z_D$ and $r_E = M\Phi(-z_D) = M - M\Phi(z_D) = M - ch - r_D$. □

The next lemma shows that work remains implementable at every continuation length $h \in (0, T]$, not only at T. Its continuity claim yields continuity of V^R in Appendix C, and its short-horizon limits enter Lemmas 4 and B.2.

Lemma A.1. *Suppose Assumption 4 holds. Then $M\overline{\Delta}(h) > ch$ for every $h \in (0, T]$; the bars $z_D(h)$ and $z_E(h)$ of (3) are continuously differentiable in h there; and r_D and r_E are continuous on $(0, T]$, with $r_D(h) \to 0$ and $r_E(h) \to M$ as $h \downarrow 0$. Consequently $h \mapsto \mathcal{A}(h)$ is continuous on $(0, T]$ in the Hausdorff metric, and $\mathcal{A}(h) \to \mathcal{A}_0$ as $h \downarrow 0$.*

Proof. Strict feasibility. Assumption 4 and Lemma 2 give $M > \underline{M}(T, c) \geq M_I(T, c)$, hence $M\overline{\Delta}(T) > cT$. Since $\overline{\Delta}(h)/h$ is strictly decreasing, as shown in the proof of Lemma 2, every $h \in (0, T)$ satisfies $\overline{\Delta}(h)/h > \overline{\Delta}(T)/T > c/M$. In particular the two roots are separated: $z_D(h) < \sqrt{h}/2 < z_E(h)$.

Continuity. Fix $h \in (0, T]$. Because $\Delta(h, \cdot)$ is strictly increasing to the left of $\sqrt{h}/2$ and strictly decreasing to its right, the partial derivative $\partial_z \Delta(h, z) = \phi(z) - \phi(z - \sqrt{h})$ is strictly positive at $z_D(h)$ and strictly negative at $z_E(h)$. Since Δ is C^1 in (h, z), the implicit function theorem applied to $\Delta(h, z) = ch/M$ makes z_D and z_E continuously differentiable at h. By Lemma 3, $r_j(h) = M\Phi\Big(z_j(h) - \sqrt{h}\Big)$ is then continuous.

Short-horizon limits. Let $h \downarrow 0$. Applying the mean value theorem to $\Delta(h, z) = \Phi(z) - \Phi(z - \sqrt{h})$ at the binding difficult bar yields $\xi_h \in \Big(z_D(h) - \sqrt{h},\ z_D(h)\Big)$ with $M\sqrt{h}\,\phi(\xi_h) = M\Delta\Big(h, z_D(h)\Big) = ch$, that is, $\phi(\xi_h) = c\sqrt{h}/M \to 0$, so $|\xi_h| \to \infty$. As $\xi_h < z_D(h) < \sqrt{h}/2 \to 0$, the family is eventually bounded above, so $\xi_h \to -\infty$; and $z_D(h) < \xi_h + \sqrt{h}$ then gives $z_D(h) \to -\infty$. Hence $r_D(h) = M\Phi\Big(z_D(h) - \sqrt{h}\Big) \leq M\Phi\Big(z_D(h)\Big) \to 0$, and by Lemma 3, $r_E(h) = M - ch - r_D(h) \to M$.

The action set. $\mathcal{A}_0 = [0, M] \times \{0\}$ does not depend on h, while $\mathcal{A}_1(h) = [r_D(h), r_E(h)] \times \{(1-c)h\}$ is a segment whose endpoints and height all vary continuously in h; Hausdorff continuity of the union follows. As $h \downarrow 0$ that segment converges to $[0, M] \times \{0\}$, which is $\mathcal{A}_0$. □

B Proofs for Section 4

Write

$$\ell_T = \frac{\phi\Big(z_D(T) - \sqrt{T}\Big)}{\phi\Big(z_D(T)\Big)} = \exp\left(z_D(T)\sqrt{T} - \frac{T}{2}\right), \qquad \lambda_T^{\mathrm{NR}} = \frac{1}{1-\ell_T}, \tag{B.1}$$

for the likelihood ratio at the no-review difficult bar and the shadow price of the no-review incentive constraint. Assumption 4 gives $z_D(T) < \sqrt{T}/2$, hence $\ell_T \in (0,1)$ and $\lambda_T^{\mathrm{NR}} > 1$.

Review-score cutoffs and incentive supply

Section 4.2 establishes that, for a given $q = Q_\lambda(x)$, the maximizing action moves through A_B, A_D, A_E, A_G at the breakpoints $q_0 < q_1 = 0 < q_2$ in (6). For every $\lambda > 0$, Q_λ is strictly increasing in x and maps $\mathbb{R}$ onto $(-\infty, \lambda - 1)$. Hence the cutoff associated with q_j is finite when $\lambda > 1 + q_j$, and inverting $Q_\lambda(k) = q_j$ gives

$$k_j(\lambda) = \frac{\tau}{2} - \log\left(1 - \frac{1+q_j}{\lambda}\right) \quad \text{if } \lambda > 1 + q_j, \qquad k_j(\lambda) = +\infty \quad \text{otherwise.} \tag{B.2}$$

For later use, define $\underline{\lambda}(h) \equiv \max\{0, 1 + q_0\}$.

Define the standardized cutoff under first-phase work by $z_j(\lambda, \tau) \equiv \Big(\tau - k_j(\lambda)\Big)/\sqrt{\tau}$. By (B.2),

$$z_j(\lambda, \tau) = \frac{\sqrt{\tau}}{2} + \frac{1}{\sqrt{\tau}} \log\left(1 - \frac{1+q_j}{\lambda}\right), \qquad j \in \{0, 1, 2\}, \tag{B.3}$$

with $z_j = -\infty$ when $k_j = +\infty$. The induced rent schedule has jumps r_D, $r_E - r_D$, and $M - r_E$ at k_0, k_1, k_2, so its first-phase incentive supply is

$$I(\lambda; \tau) = r_D\, \Delta\Big(\tau, z_0\Big) + \Big[r_E - r_D\Big]\, \Delta\Big(\tau, z_1\Big) + \Big[M - r_E\Big]\, \Delta\Big(\tau, z_2\Big), \tag{B.4}$$

where $\Delta(\tau, -\infty) = 0$. Since each tie set $\{x : Q_\lambda(x) = q_j\}$ is null, (B.4) is unaffected by how ties are resolved.

The supporting multiplier

Lemma B.1. *Fix $\tau \in (0, T)$ and suppose Assumption 4 holds.*

[(i)]

(i) Monotonicity and range. *$I(\lambda;\tau)$ is continuous and weakly increasing on $\lambda > 0$, and strictly increasing whenever the selected rent changes on a set of positive probability. Moreover,* $\lim_{\lambda\to\infty} I(\lambda;\tau) = M\overline{\Delta}(\tau)$. *If* $(1-c)h \neq r_D$, *then* $\lim_{\lambda\downarrow\underline{\lambda}(h)} I(\lambda;\tau) = 0$.

(ii) Existence. *$M\overline{\Delta}(\tau) > c\tau$, and Problem ($P_\tau$) admits a supporting multiplier $\lambda^* \geq 0$ satisfying complementary slackness.*

(iii) Positivity and binding obedience. *If $(1-c)h \neq r_D$, every supporting multiplier satisfies $\lambda^* > 0$, and pre-review obedience binds:*

$$\int_{\mathbb{R}} r^*(x)\,[g_1^\tau(x) - g_0^\tau(x)]\,dx = c\tau, \tag{B.5}$$

where r^* is the rent delivered by the optimal menu. At the knife edge $(1-c)h = r_D$, a zero multiplier may also support an optimum.

(iv) Characterization. *If $(1-c)h \neq r_D$, every supporting multiplier solves $I(\lambda;\tau) = c\tau$. The solution set is a nonempty compact interval contained in $(\underline{\lambda}(h), \infty)$, and is a singleton whenever $I(\cdot;\tau)$ is strictly increasing at that level.*

Proof. **(i)** Let $\lambda'' > \lambda'$ and let (r', s') and (r'', s'') be Lagrangian maximizers at a given review score. Write $L_\tau(x) = 1 - g_0^\tau(x)/g_1^\tau(x)$. Adding the two optimality inequalities for $s - r + \lambda L_\tau r$ gives $(\lambda'' - \lambda')L_\tau(x)\big[r''(x) - r'(x)\big] \geq 0$. Multiplying by g_1^τ and integrating yields monotonicity of I, with strictness under the stated condition. Continuity follows from (B.3): when a cutoff first becomes finite, its standardized value tends to $-\infty$ and its contribution to (B.4) vanishes.

As $\lambda \downarrow \underline{\lambda}(h)$ with $(1-c)h \neq r_D$, the assignment converges to a constant across review scores. When $1 + q_0 > 0$, it converges to A_B; when $1 + q_0 < 0$, it converges to A_D. In either case $I \to 0$. Finally, as $\lambda \to \infty$, every $k_j \to \tau/2$, so every $z_j \to \sqrt{\tau}/2$; since the three rent jumps sum to M, $I \to M\overline{\Delta}(\tau)$.

(ii) By Lemma 2 and Assumption 4, $M > \underline{M}(T,c) \geq M_I(T,c) > M_I(\tau,c) = c\tau/\overline{\Delta}(\tau)$, where the strict inequality uses $\tau < T$ and monotonicity of $M_I(\cdot,c)$. Thus pre-review obedience is strictly feasible. Slater's condition and Kuhn–Tucker, applied to the problem with $\mathrm{co}(\mathcal{A})$ in place of $\mathcal{A}$, which has the same value (Section 4.2), yield a supporting $\lambda^* \geq 0$ with complementary slackness.

(iii) Suppose $\lambda^* = 0$. Then $Q_0 \equiv -1$, so the review-score objective is $s - r$ and does not depend on x. If $(1-c)h \neq r_D$, its maximizer is unique: A_B when $(1-c)h < r_D$ and A_D when $(1-c)h > r_D$. The assigned rent is therefore constant across scores and supplies no

first-phase incentives, contradicting $c\tau > 0$. Hence $\lambda^* > 0$, and complementary slackness gives (B.5).

At $(1-c)h = r_D$, instead, $q_0 = -1$, so under $\lambda = 0$ the bad stop A_B and difficult standard A_D are tied at every review score, while A_E and A_G are strictly dominated. A score-dependent selection between the tied actions can therefore provide the required first-phase incentives, so a zero supporting multiplier cannot be ruled out.

(iv) By (iii), $\lambda^* > 0$, so binding obedience is exactly $I(\lambda;\tau) = c\tau$. Parts (i)–(ii) imply that its solution set is nonempty, compact, and bounded away from $\underline{\lambda}(h)$. Weak monotonicity makes the level set an interval, and strict monotonicity at the level makes it a singleton. □

The full ladder near the terminal date

Lemma B.2. *Suppose Assumption 4 holds and write $h = T - \tau$. Then, as $h \downarrow 0$,*

$$\frac{r_D(h)}{h} \longrightarrow +\infty, \qquad q_2(h) \longrightarrow 0, \qquad \lambda^*(T-h) \longrightarrow \lambda_T^{\mathrm{NR}} > 1,$$

with λ_T^{NR} as in (B.1), where $\lambda^*(T-h)$ denotes any supporting multiplier.

Proof. For the first limit, the difficult continuation bar satisfies $M\big[\Phi(z_D(h)) - \Phi(z_D(h) - \sqrt{h})\big] = ch$. By the mean-value theorem there is $\xi_h \in (z_D(h) - \sqrt{h}, z_D(h))$ such that $M\sqrt{h}\,\phi(\xi_h) = ch$, that is, $\phi(\xi_h) = c\sqrt{h}/M$. As shown in the proof of Lemma A.1, $\xi_h \to -\infty$, so this implies $|\xi_h| = O(\sqrt{\log(1/h)})$. Since $|z_D(h) - \sqrt{h} - \xi_h| \le \sqrt{h}$, the corresponding normal densities are asymptotically equivalent. The lower Mills bound therefore gives, for some constant $C > 0$ and all sufficiently small h, $r_D(h) = M\Phi\big(z_D(h) - \sqrt{h}\big) \ge C\sqrt{h}/|\xi_h|$. Hence $r_D(h)/h \ge C/(\sqrt{h}\,|\xi_h|) \longrightarrow +\infty$. Using $M - r_E(h) = ch + r_D(h)$, $q_2(h) = (1-c)h/\big(ch + r_D(h)\big) \longrightarrow 0$.

It remains to establish convergence of the supporting multiplier. Fix $1 < \lambda_- < \lambda_T^{\mathrm{NR}} < \lambda_+$. By Lemma A.1, $r_D(h) \to 0$ and $r_E(h) \to M$ as $h \downarrow 0$. In the incentive-supply function (B.4), the first and third terms therefore vanish, while the middle term converges to $M\Delta\big(T, z_1(\lambda, T)\big)$. At $\lambda = \lambda_T^{\mathrm{NR}}$, (B.3) and (B.1) give $z_1(\lambda_T^{\mathrm{NR}}, T) = z_D(T)$. The no-review incentive constraint therefore implies $M\Delta\big(T, z_1(\lambda_T^{\mathrm{NR}}, T)\big) = cT$. Since $z_1(\cdot, T)$ is strictly increasing and $\Delta(T, \cdot)$ is strictly increasing to the left of $\sqrt{T}/2$, the limiting incentive supply is strictly increasing in $\lambda > 1$, because $z_1(\lambda, T) < \sqrt{T}/2$ for all λ. Hence, for all sufficiently small h, $I(\lambda_-; T-h) < c(T-h) < I(\lambda_+; T-h)$. Since $r_D(h)/h \to \infty$, $(1-c)h \ne r_D(h)$ for small h, so Lemma B.1(iv) puts every supporting multiplier between λ_- and λ_+. Since the bracket can be chosen arbitrarily tightly around λ_T^{NR}, $\lambda^*(T-h) \to \lambda_T^{\mathrm{NR}}$. □

Proof of Proposition 2. By Lemma B.1, a supporting multiplier exists. Outside the knife

edge it is strictly positive, so the review-score ranking derived in Section 4.2, together with the strict monotonicity of $Q_{\lambda^*}(x)$, gives the ordered form of a^* in Proposition 2. The cutoff formula and the conditions for absent regions follow from (B.2). Since $q_0 < q_1 < q_2$, all finite cutoffs are strictly ordered.

At the knife edge $(1-c)h = r_D$, the preceding argument applies whenever the optimum is supported by a positive multiplier. If instead $\lambda^* = 0$ supports an optimum, A_B and A_D are tied at every review score, while A_E and A_G are strictly dominated. Since first-phase work is implemented, an ordered threshold selection between A_B and A_D can be chosen to satisfy obedience (by the argument of Lemma 1, a threshold selection maximizes incentive supply at a given probability of A_D). The proposition therefore continues to hold, in this case with $k_1^* = k_2^* = +\infty$. □

Proof of Corollary 1. If $k_0^* = +\infty$, the assigned rent is identically zero and cannot induce first-phase work. Hence $k_0^* < +\infty$. Since Q_{λ^*} is continuous and strictly increasing, the difficult-standard region is also nonempty. Finally, $Q_{\lambda^*}(\mathbb{R}) = (-\infty, \lambda^* - 1)$ reaches $q_1 = 0$ if and only if $\lambda^* > 1$ and reaches q_2 if and only if $\lambda^* > 1 + q_2$, which gives the three cases in the corollary. □

Proof of Proposition 3. Write $h = T - \tau$. By Lemma B.2, $\lambda^*(T-h) \to \lambda_T^{\mathrm{NR}} > 1$ and $1 + q_2(h) \to 1$. Hence, for all sufficiently small $h > 0$, $\lambda^*(T-h) > 1 + q_2(h)$. As in the proof of Lemma B.2, $(1-c)h \neq r_D(h)$ for small h, so Corollary 1 applies. Therefore $k_2^*(T-h) < +\infty$ for every sufficiently small $h > 0$, as claimed. □

C Proofs for Section 5

The fixed-review value and the one-sided derivatives

Let $u_D(h) = (1-c)h - r_D(h)$ and $u_E(h) = (1-c)h - r_E(h)$ be the principal's payoffs under the two continuation standards, with the argument suppressed where the horizon is fixed. Conditional on first-phase work, the probabilities of the four regions are determined by the standardized cutoffs, and the principal's value from the optimal fixed-τ menu is

$$\begin{aligned} V^R(\tau) = \tau &+ u_D \left[\Phi\Big(z_0(\lambda,\tau)\Big) - \Phi\Big(z_1(\lambda,\tau)\Big)\right] \\ &+ u_E \left[\Phi\Big(z_1(\lambda,\tau)\Big) - \Phi\Big(z_2(\lambda,\tau)\Big)\right] - M\Phi\Big(z_2(\lambda,\tau)\Big), \end{aligned} \tag{C.1}$$

where $\lambda = \lambda^*(\tau)$ is a supporting multiplier and $\Phi(-\infty) = 0$. Equation (C.1) applies to the two-, three-, and four-region cases under the conventions of (B.3).

For interior review dates, continuity follows from Lemma A.1 and the maximum theorem applied to the fixed-review linear program. Changes in the number of review regions may create changes in the formula used to compute the value, but not jumps in the value itself.

By Lemma 4 the review gain $G(\tau) = V^R(\tau) - V^{\mathrm{NR}}(T; c, M)$ vanishes at both boundaries, and the natural objects to study there are the one-sided derivatives

$$V_+^{R\prime}(0) := \lim_{\tau \downarrow 0} \frac{V^R(\tau) - V^{\mathrm{NR}}(T; c, M)}{\tau}, \qquad V_-^{R\prime}(T) := \lim_{\tau \uparrow T} \frac{V^R(\tau) - V^{\mathrm{NR}}(T; c, M)}{\tau - T}, \tag{C.2}$$

whenever the corresponding limits exist. These definitions do not extend V^R to either boundary.

The terminal derivative

Propositions 4(ii) and 6 turn on the sign of $V_-^{R\prime}(T)$, computed here. Throughout, $z_D(T)$ is the difficult bar of (3) at horizon T, and ℓ_T and λ_T^{NR} are defined in (B.1).

Lemma C.1. *Suppose Assumption 4 holds. Then $V_-^{R\prime}(T)$ exists, the review value and the no-review value agree to first order at T,*

$$V^R(T-h) = V^{\mathrm{NR}}(T-h; c, M) + o(h), \qquad V_-^{R\prime}(T) = \frac{\partial V^{\mathrm{NR}}(T; c, M)}{\partial T},$$

and

$$V_-^{R\prime}(T) = 1 - \lambda_T^{\mathrm{NR}} \left[c - \frac{M\phi(z_D(T) - \sqrt{T})}{2\sqrt{T}} \right]. \tag{C.3}$$

Consequently:

[(i)]

(i) $V_-^{R\prime}(T) < 0$ if and only if

$$\ell_T - (1-c) \;>\; \frac{M\phi(z_D(T) - \sqrt{T})}{2\sqrt{T}}; \tag{C.4}$$

(ii) $V_-^{R\prime}(T) < 0$ implies $\lambda_T^{\mathrm{NR}} > 1/c$, equivalently $\ell_T > 1 - c$.

Proof. Put $\tau = T - h$, and let ℓ_τ and $\lambda_\tau^{\mathrm{NR}}$ be defined by (B.1) with T replaced by τ; by Lemma A.1, $\lambda_\tau^{\mathrm{NR}} \to \lambda_T^{\mathrm{NR}} > 1$ as $h \downarrow 0$. Restricting menus to $\mathcal{A}_0$ gives the single-phase problem of length τ, whose value satisfies

$$V^{\mathrm{NR}}(\tau; c, M) = \tau - \lambda_\tau^{\mathrm{NR}} c\tau + \int_{\mathbb{R}} g_1^\tau \max\left\{0, Q_{\lambda_\tau^{\mathrm{NR}}} M\right\} dx,$$

because the maximizer pays M above the difficult bar, where $Q_{\lambda_\tau^{\mathrm{NR}}} = 0$, and obedience binds. In particular, $V^R(\tau) \geq V^{\mathrm{NR}}(\tau; c, M)$. For any $\lambda \geq 0$, adjoining obedience to the objective of (P_τ) gives $V^R(\tau) \leq \tau - \lambda c\tau + \int g_1^\tau \mathcal{V}(Q_\lambda, h)\, dx$, with $\mathcal{V}$ as in (7); at $\lambda = \lambda_\tau^{\mathrm{NR}}$,

$$0 \leq V^R(\tau) - V^{\mathrm{NR}}(\tau; c, M) \leq \int_{\mathbb{R}} g_1^\tau \Big[\mathcal{V}\Big(Q_{\lambda_\tau^{\mathrm{NR}}}, h\Big) - \max\Big\{0, Q_{\lambda_\tau^{\mathrm{NR}}} M\Big\}\Big] dx.$$

The integrand is at most $(1-c)h$ and vanishes unless $q_0 < Q_{\lambda_\tau^{\mathrm{NR}}} < q_2$. By the lower bound on r_D in the proof of Lemma B.2 and $M - r_E = ch + r_D$, this interval has width $O\Big(\sqrt{h\log(1/h)}\Big)$. Since $dQ_\lambda/dx = \lambda - 1 - Q_\lambda$ and $\lambda_\tau^{\mathrm{NR}}$ is bounded away from one, the corresponding set of review scores has probability $O\Big(\sqrt{h\log(1/h)}\Big)$, so the difference is $o(h)$. As $V^{\mathrm{NR}}(\cdot; c, M)$ is differentiable by Lemma A.1, definition (C.2) yields $V_-^{R\prime}(T) = \partial V^{\mathrm{NR}}(T; c, M)/\partial T$.

It remains to differentiate the no-review value. Implicit differentiation of $M\Big[\Phi(z_D(T)) - \Phi(z_D(T) - \sqrt{T})\Big] = cT$ gives

$$\frac{dz_D(T)}{dT} = \frac{c/M - \phi(z_D(T) - \sqrt{T})/(2\sqrt{T})}{\phi(z_D(T)) - \phi(z_D(T) - \sqrt{T})}.$$

Differentiating $V^{\mathrm{NR}}(T; c, M) = T - M\Phi(z_D(T))$ and using (B.1) gives (C.3).

For (i), substitute $\lambda_T^{\mathrm{NR}} = 1/(1-\ell_T)$ into (C.3) and multiply by $1 - \ell_T > 0$: $V_-^{R\prime}(T) < 0$ is equivalent to $c - M\phi(z_D(T) - \sqrt{T})/(2\sqrt{T}) > 1 - \ell_T$, which is (C.4). For (ii), the term $\lambda_T^{\mathrm{NR}} M\phi(z_D(T) - \sqrt{T})/(2\sqrt{T})$ in (C.3) is strictly positive, so $V_-^{R\prime}(T) < 0$ forces $1 - \lambda_T^{\mathrm{NR}} c < 0$; and $\lambda_T^{\mathrm{NR}} > 1/c$ is equivalent to $1 - \ell_T < c$. □

The left side of (C.4) is large when the reward event is weak evidence of work, that is when the shadow price of terminal incentives is high; the right side is the first-order informational gain from lengthening the horizon. Part (ii) is (C.4) with that right side replaced by zero, and hence weaker.

Corollary C.1. *Suppose $c \leq \bar{c}(T)$, so that $\underline{M}(T, c) = M_I(T, c)$. Then there exists $\delta > 0$ such that $V_-^{R\prime}(T) < 0$ for every $M \in \Big(\underline{M}(T, c), \underline{M}(T, c) + \delta\Big)$.*

Proof. Since $\underline{M}(T, c) = M_I(T, c)$, every such M satisfies Assumption 4, so Lemma C.1 applies. Let $a = \sqrt{T}/2$. As $M \downarrow M_I(T, c)$ the two no-review roots merge, so $z_D(T) \uparrow a$, $\ell_T \uparrow 1$, and $\lambda_T^{\mathrm{NR}} \to \infty$, while the bracket in (C.3) converges to $c\Big[1 - a\phi(a)/(2\Phi(a) - 1)\Big] = cF(a)/(2\Phi(a) - 1)$, which is strictly positive because $F(a) > 0$ (proof of Lemma 2). Hence $V_-^{R\prime}(T) \to -\infty$, which gives the claim. By contrast, as $M \to \infty$, $\ell_T \to 0$ and $M\phi(z_D(T) - \sqrt{T}) = \ell_T\, M\phi(z_D(T)) \to 0$, so $V_-^{R\prime}(T) \to 1 - c > 0$, consistent with Proposition 5. □

Proof of Lemma 4. Consider first $\tau \downarrow 0$. The largest principal payoff available from a second-phase action is $u_D = (1-c)h - r_D$ for all sufficiently small τ, because profitability of the

no-review contract implies $u_D(T) = V^{\mathrm{NR}}(T; c, M) > 0$. Hence $V^R(\tau) \le \tau + u_D(T-\tau)$, whose limit is $u_D(T) = V^{\mathrm{NR}}(T; c, M)$.

For a matching lower bound, assign the low-rent continuation action A_D at every review score except on a sufficiently high upper tail, on which the principal assigns the good stop A_G. The utility increment on that tail is $M - r_D$. Choose the upper-tail cutoff so that $\big[M - r_D\big]\Delta(\tau, z) = c\tau$. For sufficiently small τ, such a cutoff exists, and the work probability of the reward region converges to zero. The resulting payoff converges to $u_D(T)$. Therefore $\lim_{\tau\downarrow 0} V^R(\tau) = V^{\mathrm{NR}}(T; c, M)$.

Now let $\tau \uparrow T$, so $h \downarrow 0$. A menu using only A_B and A_G is exactly a bounded upper-tail bonus based on X_τ. Choosing its threshold to satisfy $M\,\Delta(\tau, z) = c\tau$ gives payoff $V^{\mathrm{NR}}(\tau; c, M)$, which converges to $V^{\mathrm{NR}}(T; c, M)$.

For the reverse inequality, Lemma C.1 gives $V^R(T-h) \le V^{\mathrm{NR}}(T-h; c, M) + o(h)$, and $V^{\mathrm{NR}}(\cdot; c, M)$ is continuous by Lemma A.1. This proves $\lim_{\tau\uparrow T} V^R(\tau) = V^{\mathrm{NR}}(T; c, M)$. □

Proof of Proposition 4. The two claims concern the one-sided derivatives (C.2). Part (i) establishes that $V_+^{R\prime}(0) = -\infty$ directly; part (ii) reads the sign of $V_-^{R\prime}(T)$ off Lemma C.1.

For the first claim, write $h = T - \tau$ and use the low-rent continuation action $A_D(h)$ as a baseline. For all sufficiently small τ, this action uniquely maximizes the principal's post-review payoff. Compactness of the two continuation frontiers and strict profitability at $h = T$ imply that there is a constant $\kappa > 0$ such that every feasible review action (r, s) satisfies $u_D - (s - r) \ge \kappa|r - r_D|$. Let $L_\tau(x) = 1 - g_0^\tau(x)/g_1^\tau(x)$. Because a constant review utility provides no incentives, pre-review obedience can be written as $\mathbb{E}_1\left[\big(r(X_\tau) - r_D\big)L_\tau(X_\tau)\right] \ge c\tau$. Under work, $X_\tau = \tau + \sqrt{\tau}Z$ and $L_\tau(X_\tau) = 1 - \exp\big(-\tau/2 - \sqrt{\tau}Z\big)$. Split at $|Z| = \sqrt{8\log(1/\tau)}$. On the central event, $|L_\tau|$ is of order $\sqrt{\tau\log(1/\tau)}$; under both work and shirking, the complementary Gaussian tails contribute $o(\tau)$. Since review utilities are bounded by M, obedience therefore implies $\mathbb{E}_1\big[|r(X_\tau) - r_D|\big] \ge K\sqrt{\tau}/\sqrt{\log(1/\tau)}$ for some $K > 0$ and all sufficiently small τ. Combining this with $u_D - (s-r) \ge \kappa|r - r_D|$ yields $V^R(\tau) \le \tau + u_D(T-\tau) - \kappa K\sqrt{\tau}/\sqrt{\log(1/\tau)}$. Since $\tau + u_D(T-\tau) = V^{\mathrm{NR}}(T; c, M) + O(\tau)$, it follows that

$$\lim_{\tau\downarrow 0} \frac{V^R(\tau) - V^{\mathrm{NR}}(T; c, M)}{\tau} = -\infty. \tag{C.5}$$

Thus $V^R(\tau) < V^{\mathrm{NR}}(T; c, M)$ for all sufficiently small $\tau > 0$. Since the boundary limit established above gives $V^R(s) \to V^{\mathrm{NR}}(T; c, M)$ as $s \downarrow 0$, each such τ is strictly dominated by some genuine review date $s \in (0, \tau)$. This proves part (i).

For the second claim, let M lie in the right neighborhood supplied by Corollary C.1, so that $V_-^{R\prime}(T) < 0$. By Lemma C.1, $V^R(T-h) = V^{\mathrm{NR}}(T; c, M) - V_-^{R\prime}(T)h + o(h)$. Comparing

the dates $\tau = T - h$ and $\tau' = T - 2h$, $V^R(\tau') - V^R(\tau) = -V_-^{R\prime}(T)h + o(h) > 0$ for all sufficiently small $h > 0$. Hence every sufficiently late review is strictly dominated by an earlier genuine review, which proves the second claim. □

Proof of Proposition 5. Write $R_M(t) \equiv r_D(t)$ for the minimal work-inducing rent over a phase of length t. By Proposition 1, $R_M(T) \to 0$ as $M \to \infty$, so the no-review full-effort contract is feasible and profitable for all sufficiently large M. Suppose toward a contradiction that there are sequences $M_n \to \infty$ and $\tau_n \in (0, T)$ such that

$$V^R(\tau_n) \geq V^{\mathrm{NR}}(T; c, M_n). \tag{C.6}$$

Dependence on n is suppressed below. Any continuation-utility schedule $r(X_\tau) \in [0, M]$ that induces first-phase work is itself a bounded incentive contract for a phase of length τ. By the definition of $R_M(\tau)$, the agent's ex ante rent under the review contract is therefore at least $R_M(\tau)$. Total surplus is at most $(1-c)T$, and hence $V^R(\tau) \leq (1-c)T - R_M(\tau)$. Comparison with (C.6) yields the necessary condition $R_M(\tau) \leq R_M(T)$.

We first show that any such sequence must satisfy $\tau_n \log M_n \longrightarrow 0$. Implicit differentiation of the binding difficult-root condition gives

$$R_M'(t) = \frac{\phi(z_D(t) - \sqrt{t})}{\phi(z_D(t)) - \phi(z_D(t) - \sqrt{t})} \left[c - \frac{M\phi(z_D(t))}{2\sqrt{t}} \right]. \tag{C.7}$$

For every $\varepsilon > 0$, $z_D(t) \to -\infty$ uniformly on $t \in [\varepsilon, T]$ as $M \to \infty$. The Mills bound $\Phi(z) \leq \phi(z)/|z|$ for $z < 0$, together with $M\Delta(t, z_D(t)) = ct$, makes the bracket in (C.7) negative uniformly on that interval. The prefactor is positive on the difficult branch. Hence $R_M'(t) < 0$ on $[\varepsilon, T]$ for all sufficiently large M. Condition $R_M(\tau) \leq R_M(T)$ then forces $\tau_n \to 0$.

We sharpen this localization. Let $a_M < 0$ solve $\Phi(a_M) = 2cT/M$, and put $q_M = \Phi(a_M - \sqrt{T})/\Phi(a_M)$. The threshold bonus $M/[2(1-q_M)]$ is feasible for large M, implements effort, and leaves rent $cTq_M/(1-q_M)$. Since Φ/ϕ is increasing, $q_M \leq \exp(a_M\sqrt{T} - T/2)$, and $|a_M| \geq \sqrt{\log M}$ for large M; therefore

$$R_M(T) \leq 2cT \exp\left(-\sqrt{T \log M} - \frac{T}{2} \right) \tag{C.8}$$

for all sufficiently large M.

For a short phase of length τ, write $z_D(\tau) = -x$ and $u = \sqrt{\tau}$. The binding condition implies $R_M(\tau) = c\tau q/(1-q) \geq c\tau q$, where $q = \Phi(-(x+u))/\Phi(-x)$. The two-sided Mills

inequalities and $x \le \sqrt{2\log(M/(c\tau))}$ give

$$R_M(\tau) \ge \frac{c\tau}{2(1+\sqrt{T})} \exp\left[-\sqrt{2\tau \log\left(\frac{M}{c\tau}\right)} - \frac{\tau}{2}\right]. \tag{C.9}$$

If $\tau_n \log M_n$ were bounded away from zero along a subsequence, then the logarithm of the right side of (C.9) would be $-o(\sqrt{\log M_n})$, whereas (C.8) is of order $-\sqrt{T \log M_n}$. It would follow that $R_{M_n}(\tau_n) > R_{M_n}(T)$, contradicting $R_M(\tau) \le R_M(T)$. This proves $\tau_n \log M_n \longrightarrow 0$.

It remains to show that a review in this vanishingly early region is also strictly dominated. Put $h = T - \tau$, $r_0 = R_M(h)$, and let the difficult continuation action be (r_0, s_0), where $s_0 = (1-c)h$ is total surplus and its principal payoff is $s_0 - r_0$. Uniformly for $h \in [T/2, T]$, $r_0 \to 0$; hence $2r_0 < (1-c)h$ for all sufficiently large M. Every continuation action $(r, s) \in \mathcal{A}(h)$ then satisfies $s - r \le s_0 - r_0 - |r - r_0|$. For a work-inducing action $s = (1-c)h = s_0$ and $r \ge r_0$, so the inequality holds with equality. For a no-effort action $s = 0$; it follows immediately when $r \ge r_0$, and when $r < r_0$ it follows from $2r_0 < (1-c)h$.

Let $d(x) = r(x) - r_0$. Integrating $s - r \le s_0 - r_0 - |r - r_0|$ under first-phase work gives

$$V^R(\tau) \le \tau + s_0 - r_0 - \mathbb{E}_1|d(X_\tau)|. \tag{C.10}$$

Since a constant rent contributes nothing to incentives, first-phase obedience is, with L_τ as in the proof of Proposition 4,

$$\mathbb{E}_1[d(X_\tau)L_\tau(X_\tau)] \ge c\tau. \tag{C.11}$$

Under work, $X_\tau = \tau + \sqrt{\tau}Z$ and $L_\tau(X_\tau) = 1 - \exp(-\tau/2 - \sqrt{\tau}Z)$. On the event $|L_\tau| \le 1/2$, the left side of (C.11) is bounded by $\mathbb{E}_1|d|/2$. On its complement, $|d| \le M$, and Gaussian tail bounds give, with $a = \log(3/2)$, $\mathbb{E}_1\left[|L_\tau|\mathbf{1}_{\{|L_\tau|>1/2\}}\right] \le 4\exp\left(-a^2/(8\tau)\right)$. Because $\tau_n \log M_n \longrightarrow 0$ is equivalent to $\log M = o(1/\tau)$, this tail contribution is $o(\tau)$. Thus

$$\mathbb{E}_1|d(X_\tau)| \ge 2c\tau - o(\tau) > c\tau \tag{C.12}$$

for all sufficiently large n.

Finally, subtracting $V^{\mathrm{NR}}(T; c, M) = (1-c)T - R_M(T)$ from (C.10) yields

$$V^R(\tau) - V^{\mathrm{NR}}(T; c, M) \le c\tau + R_M(T) - R_M(T-\tau) - \mathbb{E}_1|d(X_\tau)|.$$

For large n, $T - \tau_n \in [T/2, T)$ and the monotonicity established after (C.7) gives $R_M(T-\tau) > R_M(T)$. Combining this strict inequality with (C.12) makes the displayed difference negative, contradicting (C.6). No such sequence exists, so a finite uniform $\overline{M}(T, c)$ exists. □

Proof of Proposition 6. By Lemma 4, $G(\tau) \to 0$ as $\tau \uparrow T$, and $V_-^{R\prime}(T)$ in (C.2) is exactly the left derivative of G at T: $G(\tau) = V_-^{R\prime}(T)\,(\tau - T) + o(T - \tau)$ as $\tau \uparrow T$. Under the stated hypotheses, Corollary C.1 gives $V_-^{R\prime}(T) < 0$. Since $\tau - T < 0$, the leading term is strictly positive, so there is $\delta > 0$ with $G(\tau) > 0$ for every $\tau \in (T - \delta, T)$. □

Proof of Proposition 7. At the boundary, $\overline{\Delta}(T)/T = c/M_0$. Since $\overline{\Delta}(t)/t$ is strictly decreasing by Lemma 2, every $t < T$ satisfies $\overline{\Delta}(t)/t > c/M_0$, which gives $M_0\overline{\Delta}(\tau) > c\tau$ and $M_0\overline{\Delta}(h) > ch$.

To obtain the second claim, choose $\tau_0 > 0$ small and write $h_0 = T - \tau_0$. At $M = M_0$ the continuation problem at $h_0 < T$ is strictly feasible, so its minimum work-inducing rent $r_D(h_0)$ is well defined and strictly below M_0. At the boundary $h = T$, the two threshold roots coincide and the corresponding rent is $M_0\Phi(-\sqrt{T}/2)$; hence the available rent spread converges to $M_0\Phi(\sqrt{T}/2) > 0$ as $h_0 \uparrow T$. Since $\overline{\Delta}(\tau)/\tau \to \infty$ as $\tau \downarrow 0$, τ_0 can therefore be chosen small enough that $\big[M_0 - r_D(h_0)\big]\overline{\Delta}(\tau_0) > c\tau_0$. Both continuation feasibility at h_0 and the above inequality are strict, so by continuity they continue to hold for all caps in a sufficiently small interval $(M_0 - \varepsilon, M_0]$.

For such a cap, use the minimum-rent full-effort continuation contract after review scores $X_{\tau_0} < \tau_0/2$, and use the good-stop action with rent M after $X_{\tau_0} \geq \tau_0/2$. The spread in continuation utility between the two regions is $M - r_D(h_0)$, and the midpoint cutoff maximizes the difference between the work and shirk probabilities. The date-0 incentive spread is therefore $\big[M - r_D(h_0)\big]\overline{\Delta}(\tau_0) > c\tau_0$, so first-phase work is induced. In the lower region the continuation contract induces full effort, while in the upper region the project is stopped. The lower region has strictly positive probability under work. Finally, $M < M_0 = M_I(T, c)$ makes full effort infeasible without review by the implementability condition in Lemma 2. □

D Proofs for Section 6

Proof of Proposition 8. Let $Y \in [0, M]$ denote the terminal payment and $W_t = \mathbb{E}_t[Y - c\int_t^T e_s\,ds]$ promised utility. Martingale representation gives $dW_t = ce_t\,dt + \beta_t(dX_t - e_t\,dt)$ with $W_T = Y$. A local deviation from work to shirk saves c and reduces the drift of promised utility by β_t, so work requires $\beta_t \geq c$. Also $0 \leq W_t \leq M$: permanent shirking guarantees a nonnegative continuation payoff, whereas $Y \leq M$. At $W_t = 0$, incentive compatibility forces both the terminal payment and all subsequent effort to be zero. At $W_t = M$, the identity $M = \mathbb{E}_t Y - c\mathbb{E}_t\int_t^T e_s ds \leq M - c\mathbb{E}_t\int_t^T e_s ds$ gives the same conclusion, with terminal payment M. Thus both boundaries are absorbing.

Scale promises by c: $u = W/c$, $m = M/c$, and let $v(h, u)$ be maximal expected remaining

work. Because $\mathbb{E}Y = W_0 + c\mathbb{E}\int_0^T e_t dt$, the principal's payoff is $(1-c)\mathbb{E}\int_0^T e_t dt - W_0$. With $b = \beta/c$, the recursive equation is

$$v_h = \max\left\{\sup_{b\geq 1}[1 + v_u + \tfrac{1}{2}b^2 v_{uu}], \sup_{b\leq 1}\tfrac{1}{2}b^2 v_{uu}\right\},$$

with $v(h,0) = v(h,m) = v(0,u) = 0$. Under the candidate policy the agent works in the interior with $b = 1$, so $U_t = W_t/c$ satisfies $dU_t = dt + dB_t$ until its first exit σ from $(0,m)$ and $v(h,u) = \mathbb{E}_u[\sigma \wedge h]$. Hence $v_h = 1 + v_u + \frac{1}{2}v_{uu}$. The parabolic maximum principle applied to v_h and v_{uu} gives $v_h \geq 0$ and $v_{uu} \leq 0$. The work branch is therefore maximized at the smallest admissible b, namely 1, while the shirk branch is at most zero and cannot exceed v_h. Verification yields the band policy. Returning to dollars gives $dW_t = c\,dX_t$, hence $W_t = W_0 + cX_t$ until a boundary is hit. Finally the principal chooses $u = W_0/c$ to maximize $(1-c)v(T,u) - cu$; strict concavity of v gives uniqueness whenever the optimum is interior. □

Proof of Proposition 9. Before τ_1 the agent learns nothing about the realized review date, so a deviation is summarized by cumulative effort $e \in [0, \tau_1]$. If the early review occurs his payoff is $U_1(e)$; otherwise he learns that the date is τ_2 and reoptimizes over cumulative effort in $[e, e+d]$, obtaining $\widehat{U}_2(e)$. Sequential optimality after non-arrival therefore requires $U_2(\tau_2) = \widehat{U}_2(\tau_1)$. Given this, full effort before τ_1 is optimal if and only if $pU_1(\tau_1) + (1-p)U_2(\tau_2) \geq pU_1(e) + (1-p)\widehat{U}_2(e)$ for every e, which is (12). This proves part (i).

If the late branch induces full effort on its own, then $D_2(e) \geq 0$. Where $D_1(e) \geq 0$ the constraint holds for every p; where $D_1(e) < 0$ it is equivalent to $p \leq D_2(e)/[D_2(e) - D_1(e)]$. Taking the infimum gives (13), proving part (ii). Under full effort the lottery payoff is $P_2 + p(P_1 - P_2)$. When $P_1 > V^{D,\mathrm{pw}} > P_2$, this exceeds $V^{D,\mathrm{pw}}$ if and only if $p > \underline{p}$, whereas sequential incentive compatibility requires $p \leq \overline{p}$. Thus a feasible strict improvement exists if and only if $\underline{p} < \overline{p}$, and every p in the stated interval works, proving part (iii). □

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Online Appendix to "Midterm Review"

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OA.1 Pointwise effort choice

Section 6.1 states the two conclusions of this extension: no interior effort level becomes an extreme point of the continuation action set, and the easy continuation standard must be raised from $z_E(h)$ to $\zeta(h)$, lowering the maximum continuation rent from $r_E(h)$ to $\widetilde{r}_E(h)$. This appendix supplies the argument.

OA.1.1 Flexible effort and a relaxed feasible set

Fix a continuation phase of length h and let total effort be $\eta \in [0, h]$, so that a plan with total effort η produces $Y \sim N(\eta, h)$ and costs the agent $c\eta$. For a terminal wage w write $\pi(\eta) = \mathbb{E}_\eta[w] = \int_{\mathbb{R}} w(y) f_\eta(y)\, dy$, where f_η is the $N(\eta, h)$ density, for the agent's expected payment when he supplies total effort η, and let $\mathcal{F}_h^{\text{flex}}(\eta)$ denote the contracts satisfying ($\text{IC}_h^{\text{flex}}$). Proofs are collected in Section OA.1.6.

The economic difference from phase-level effort is that the old model asks only whether the agent wants to work rather than not work, whereas the flexible model also asks whether he wants to work *all the way to the target*. Those nearby deviations are what tighten the easy standard below.

A relaxed feasible set. Condition ($\text{IC}_h^{\text{flex}}$) contains a continuum of deviations, and we will not need to characterize all of them. Three necessary conditions suffice. Two are already in the main text: (G_η), the comparison with abandoning effort altogether, which is exactly the comparison used in the phase-level model; and (L_η), which, because $\pi'(\eta) = \text{Cov}_\eta(w(Y), Y)/h$, says that the wage must load enough on output to pay for the *marginal* unit of effort. The second is the new restriction that matters for the easy standard. Third, at an interior target the agent's payoff must bend the right way:

$$\pi''(\eta) \leq 0 \qquad \text{if } \eta \in (0, h). \tag{S_η}$$

Let $\mathcal{F}_h(\eta)$ impose only (G_η), and let $\widehat{\mathcal{F}}_h(\eta)$ impose the global, local, and second-order conditions. Then

$$\mathcal{F}_h^{\text{flex}}(\eta) \;\subseteq\; \widehat{\mathcal{F}}_h(\eta) \;\subseteq\; \mathcal{F}_h(\eta). \tag{OA.1}$$

The rightmost set is the old binary-effort benchmark. The middle set is an analytically useful outer bound on the true flexible-effort set. The strategy below is to show that even this larger set has no economically relevant extreme points beyond the four actions we identify. Since the four candidate vertices themselves satisfy the full incentive constraint, this sandwich argument is enough; we never have to solve the entire continuum-deviation problem at interior effort.

Rent and surplus. As in the baseline model, an action is the pair (r, s) of continuation rent and *total* continuation surplus, the principal's own payoff being $u = s - r$. If the contract implements η,

$$r = \pi(\eta) - c\eta, \qquad s = (1-c)\eta, \qquad u = s - r = \eta - \pi(\eta). \tag{OA.2}$$

For a fixed effort target, feasible contracts therefore lie on a horizontal segment at height $(1-c)\eta$, along which the principal's payoff falls one for one with the rent conceded. Flexible effort fills in all the segments corresponding to $\eta \in [0, h]$:

$$\mathcal{A}^{\text{flex}}(h) = \bigcup_{\eta \in [0,h]} \Big\{ \Big(P - c\eta,\ (1-c)\eta \Big) :\ P \in [P_{\min}(\eta), P_{\max}(\eta)] \Big\}, \tag{OA.3}$$

where $P_{\min}(\eta)$ and $P_{\max}(\eta)$ are the least and greatest expected payments that can implement η. Define $\widehat{P}_{\min}$ and $\widehat{P}_{\max}$ analogously using the relaxed set $\widehat{\mathcal{F}}_h(\eta)$.

The picture is therefore richer than in the baseline model: instead of two line segments, the principal faces a two-dimensional body. The key question, however, is not how large that body is. It is whether the body creates new *extreme* rent–surplus pairs. Since the review problem maximizes a linear function of (r, s) score by score, only those extreme points can ever matter for the optimal menu.

OA.1.2 The three regimes

There are three economically distinct targets: partial effort, full effort, and no effort. Partial effort is the new case. Full effort turns out to look almost exactly like the baseline model, while no effort is unchanged.

Interior effort: why a band replaces a threshold. Suppose the principal wants to implement $\eta \in (0, h)$. In the baseline model an upper-tail bonus is attractive because high output is evidence of effort. With an interior target that logic is incomplete: a very high

outcome is also evidence that the agent supplied *more* than the target. A contract that keeps paying on arbitrarily high outcomes therefore encourages the wrong upward deviation.

The least-cost way to induce an interior target consequently rewards a *range* of good outcomes rather than the entire upper tail. The lower edge of the range makes effort attractive; the upper edge prevents the agent from wanting to overshoot the target. Formally, matching both the expected payment and the marginal incentive produces the following band-reduction result.

Lemma OA.1. *Fix $\eta \in (0,h)$ and let $w:\mathbb{R}\to[0,M]$ satisfy $\pi'_w(\eta)=c$. Then there is a band bonus*

$$\tilde{w}(y) = M\mathbf{1}\{k_1 \le y \le k_2\}, \qquad -\infty \le k_1 < k_2 \le +\infty, \tag{OA.4}$$

with the same expected payment and the same marginal incentive at the target,

$$\pi_{\tilde{w}}(\eta) = \pi_w(\eta), \qquad \pi'_{\tilde{w}}(\eta) = c, \qquad \text{and} \qquad \pi_{\tilde{w}}(0) \le \pi_w(0).$$

Consequently $\tilde{w}$ satisfies (G_η) and (L_η) whenever w does, so $\widehat{P}_{\min}(\eta)$ is bounded below by the least payment among band bonuses satisfying these two conditions.

The economic content is simple. The principal still wants to spend the payment cap where output is informative about effort. But she now has to control incentives on both sides of the target. The optimal payment region is therefore bounded above as well as below: outcomes that are sufficiently good earn the bonus, while extremely good outcomes do not. The upper cutoff is not a punishment for success; it is what prevents the contract from rewarding effort beyond the amount the principal wants to buy.

Lemma OA.1 reduces the problem to two numbers. Standardize the band by

$$a = \frac{k_1-\eta}{\sqrt{h}}, \qquad b = \frac{k_2-\eta}{\sqrt{h}}, \qquad \kappa = \frac{c\sqrt{h}}{M},$$

so that $\pi(\eta) = M[\Phi(b)-\Phi(a)]$ and $\pi'(\eta) = M[\phi(a)-\phi(b)]/\sqrt{h}$. The interior condition in (L_η) then reads

$$\phi(a) - \phi(b) = \kappa, \tag{OA.5}$$

a single equation in (a,b) that does not involve η. Let $\zeta(h)>0$ be the bar of Section 6.1, the positive root of $\phi(\zeta)=\kappa$, equivalently of $M\phi(\zeta)/\sqrt{h}=c$. Along the curve (OA.5) the payment $M[\Phi(b)-\Phi(a)]$ is largest at $a=-\zeta(h)$, $b=+\infty$: the band degenerates to an upper-tail bonus with bar $\eta-\zeta(h)\sqrt{h}$, and the payment it delivers,

$$p^* = M\,\Phi\Big(\zeta(h)\Big), \tag{OA.6}$$

uses (L_η) alone and does not vary with η. The bound is attained, and the slice is an interval.

Lemma OA.2. *Suppose* $\Delta\big(h, \zeta(h)\big) \geq ch/M$. *Then for every* $\eta \in (0, h)$ *the expected payments achievable at* η *form an interval with upper end* p^* *and lower end at least* $\widehat{P}_{\min}(\eta)$. *The upper end is attained by the upper-tail bonus with bar* $\eta - \zeta(h)\sqrt{h}$, *which lies in* $\mathcal{F}_h^{\text{flex}}(\eta)$ *itself and not merely in the relaxation. Here* $\widehat{P}_{\min}(\eta) = M[\Phi(b) - \Phi(a)]$ *at the solution of* (OA.5) *together with* (G_η) *at equality: writing* $\theta = \eta/\sqrt{h}$,

$$\Big[\Phi(b) - \Phi(a)\Big] - \Big[\Phi(b+\theta) - \Phi(a+\theta)\Big] = \kappa\theta. \tag{OA.7}$$

Because the maximum expected payment p^* is the same for every interior target, the high-rent side of the feasible set has a particularly simple interpretation. Along that side, the principal is already paying as much as the local incentive condition permits. Increasing the effort target therefore means asking the agent to work more without increasing expected compensation. Substituting $P = p^*$ into (OA.2) gives $(r, s) = \big(p^* - c\eta,\ (1-c)\eta\big)$ for $\eta \in [0, h]$, which runs from the no-effort face to

$$\widetilde{A}_E = \big(p^* - ch,\ (1-c)h\big) \tag{OA.8}$$

at full effort.

This already suggests why partial effort will not survive in the final menu. Every interior point on this high-rent locus can be replicated by mixing the full-effort endpoint $\widetilde{A}_E$ with no-effort actions. The only candidate new extreme action is therefore the full-effort endpoint itself. Economically, flexible effort does not create a new kind of reward; it merely limits how generous the easy full-effort reward can be.

Full effort. At full effort the agent cannot overshoot the target. The relevant contract is therefore again an upper-tail bonus, exactly as in the baseline model. What changes is that full effort must now beat *every* lower effort level, not just zero effort. For an upper-tail bonus this creates two transparent tests: a global test comparing full effort with zero effort and a marginal test asking whether the last unit of effort is worth its cost. Remarkably, for the Gaussian threshold contract these two tests are jointly sufficient.

Lemma OA.3. *Let* $w = M\mathbf{1}\{y \geq h - z\sqrt{h}\}$. *Then* w *implements* $\eta = h$ *in the sense of* (IC_h^{flex}) *if and only if*

$$\Delta(h, z) \geq \frac{ch}{M} \qquad \textit{and} \qquad \phi(z) \geq \kappa.$$

The two inequalities measure different ways in which the same standard can fail. The first asks whether the standard generates enough incentive *on average* to make full effort

preferable to giving up altogether. The second asks whether it generates enough incentive *at the margin* to stop the agent from shaving the last bit of effort. Phase-level effort checks only the first. Flexible effort checks both.

Lemma OA.4. *Write* $\Lambda(h,z) = \sqrt{h}\,\phi(z)$*, let* $z^\dagger(h)$ *be the unique bar at which* $\Delta(h,z) = \Lambda(h,z)$ *— it lies in* $\big(0, \sqrt{h}/2\big)$ *— and set*

$$\overline{\mathcal{M}}(h) = \max_z \ \min\Big\{\Delta(h,z),\ \Lambda(h,z)\Big\} = \sqrt{h}\,\phi\Big(z^\dagger(h)\Big). \tag{OA.9}$$

(i) Full effort is implementable under flexible effort if and only if $\overline{\mathcal{M}}(h) \geq ch/M$*; and* $\overline{\mathcal{M}}(h) < \overline{\Delta}(h)$ *for every* $h > 0$*, so flexible effort always tightens feasibility.*

(ii) When that inequality is strict, the full-effort slice of $\mathcal{A}^{\text{flex}}(h)$ *is the segment of the line* $s = (1-c)h$ *joining* A_D *to* $\widetilde{A}_E$*.* The difficult end is unchanged*: it is borne by the bar* $z_D(h)$ *of Section 3.1, at which the global condition binds and the local one is slack, so*

$$r_D(h) = M\,\Phi\Big(z_D(h) - \sqrt{h}\Big), \qquad \pi'(h) = \frac{M\,\phi\Big(z_D(h)\Big)}{\sqrt{h}} > c.$$

The easy end changes*: it is borne by the strictly harder bar* $\zeta(h) < z_E(h)$*, at which the local condition binds and the global one is slack, so*

$$\widetilde{r}_E(h) = M\,\Phi\Big(\zeta(h)\Big) - ch \ < \ r_E(h), \qquad \pi'(h) = c. \tag{OA.10}$$

Which of the two conditions of Lemma OA.3 binds is what makes the correction asymmetric, for the reason given in Section 6.1: the difficult standard is already sensitive to effort at the margin, whereas the easy one is not. The magnitudes are worth recording. Feasibility itself changes only modestly: $\overline{\Delta}(h)/\overline{\mathcal{M}}(h) = 1 + h/72 + O(h^2)$. The larger effect is on the rent that the easy action can promise. For example, at $h = 1/4$ and $c = 0.3$, $\widetilde{r}_E/r_E$ ranges from about 0.84 near the feasibility floor to 0.98 when the cap is four times that floor; at $h = 1$ the ratio can fall to about 0.68 near the tight end. Flexible effort therefore matters most exactly where continuation rents are scarce and valuable as incentive capital.

No effort. Nothing changes at the no-effort end. A constant payment $w \equiv p$ gives the same expected payment at every effort level, so exerting positive effort only adds cost. Hence every $p \in [0, M]$ implements zero effort and delivers $(r, s) = (p, 0)$. The entire old no-effort segment remains feasible, including $A_B = (0,0)$ and $A_G = (M, 0)$.

OA.1.3 Extreme points

The review problem cares not about all feasible contracts but about the convex hull of feasible rent–surplus pairs, which is what Figure 4 draws. This section supplies the characterization behind that figure: the relaxed pointwise-effort set lies inside $\mathrm{co}\{A_B, A_D, \widetilde{A}_E, A_G\}$, with $\widetilde{A}_E$ as in (OA.8) and A_B, A_D, A_G as in Proposition 2, and each of those four vertices is implementable. The one non-obvious step is the low-rent boundary: partial effort never delivers effort more cheaply in rent-per-unit terms than full effort.

The residual question, in rent terms. Could the principal ever prefer to buy *some* continuation effort rather than the full amount, because partial effort requires less rent per unit? Let $\rho(\eta)$ denote the rent under the cheapest relaxed contract implementing η:

$$\rho(\eta) = \widehat{P}_{\min}(\eta) - c\eta, \qquad \text{with} \qquad \rho(h) = M\Phi\Big(z_D(h)\Big) - ch = r_D(h). \tag{OA.11}$$

The relevant chord inequality is equivalent to

$$\frac{\rho(\eta)}{\eta} \;\geq\; \frac{\rho(h)}{h} = \frac{r_D(h)}{h} \qquad \text{for every } \eta \in (0,h). \tag{OA.12}$$

So the issue has a direct economic interpretation: is full effort the cheapest way to buy effort in rent terms? The answer is yes. Rent per unit falls as the principal buys more effort, so a partial-effort continuation is never an extreme point.

The normalization used in the proof makes this a one-dimensional statement. Writing $\theta = \eta/\sqrt{h}$, the minimum rent takes the form

$$\rho(\eta) = M\,\mathcal{R}_\kappa\Big(\eta/\sqrt{h}\Big), \qquad \mathcal{R}_\kappa(\theta) = \Phi(b+\theta) - \Phi(a+\theta), \tag{OA.13}$$

where (a,b) solve the band conditions. The next lemma establishes the desired economies of scale in incentive rent.

Lemma OA.5. *Suppose $\overline{\mathcal{M}}(h) > ch/M$. Then*

$$\widehat{P}_{\min}(\eta) \;\geq\; \frac{\eta}{h}\,\widehat{P}_{\min}(h) \qquad \textit{for every } \eta \in (0,h);$$

equivalently, by (OA.11), the agent's rent per unit of effort implemented, $\rho(\eta)/\eta$, is smallest at full effort.

Proposition OA.1. *Suppose $\overline{\mathcal{M}}(h) > ch/M$. Then*

$$\mathrm{co}\,\mathcal{A}^{\mathrm{flex}}(h) = \mathrm{co}\Big\{A_B,\ A_D,\ \widetilde{A}_E,\ A_G\Big\},$$

and all four vertices belong to $\mathcal{A}^{\text{flex}}(h)$*. In particular, for every* $q \in \mathbb{R}$ *a maximizer of* $s + qr$ *over* $\mathcal{A}^{\text{flex}}(h)$ *can be chosen from these four actions, and no interior effort level is ever used.*

This is the payoff from the preceding machinery. Although flexible effort creates a continuum of possible effort levels and a two-dimensional feasible set, none of the interior effort levels is ever chosen by the principal. The economically relevant set still has four vertices. Three are exactly the baseline actions, and the fourth is the same easy continuation with a slightly harder standard.

The proof deliberately uses the relaxed set only as an outer bound. We therefore do not need a complete characterization of incentive compatibility at every interior target.

OA.1.4 Consequences for the analysis

Proposition OA.1 is what licenses the substitution $r_E(h) \mapsto \widetilde{r}_E(h)$ announced in Section 6.1. Only the upper breakpoint of (6) moves, to $\widetilde{q}_2(h) = (1-c)h/\Big(M - \widetilde{r}_E(h)\Big) < q_2(h)$, so the good-stop region opens at a lower shadow value than in the baseline, while the cutoff formula (B.2) is unchanged and the ordered four-region menu of Proposition 2 survives verbatim. Feasibility of full continuation effort is now the strictly stronger requirement $\overline{\mathcal{M}}(h) \geq ch/M$ of Lemma OA.4(i), in place of $\overline{\Delta}(h) \geq ch/M$.

OA.1.5 Flexible effort before the review

So far we have allowed the agent to coast only after the review. The same concern can arise before the review: an agent who is supposed to work until τ may instead work for only part of that interval. The effect is again local and economically transparent. Because the desired first-phase effort is the upper boundary $\eta_1 = \tau$, the principal does not need to make the agent indifferent at the margin; she only needs the marginal return to the last bit of effort to be at least its cost.

Only one new constraint. Let $\eta_1 \in [0, \tau]$ denote total first-phase effort. The review score is $X_\tau \sim N(\eta_1, \tau)$, and the menu promises continuation utility $r(x)$ after review score x. Implementing full first-phase effort requires $\mathbb{E}_\tau[r(X_\tau)] - c\tau \geq \mathbb{E}_{\eta_1'}[r(X_\tau)] - c\eta_1'$ for every $\eta_1' \in [0, \tau]$. The baseline model checks only the most distant deviation, $\eta_1' = 0$. Flexible effort adds the possibility of shaving a small amount of first-phase effort. At the full-effort corner this requires

$$\frac{d}{d\eta_1}\mathbb{E}_{\eta_1}\Big[r(X_\tau)\Big]\Big|_{\eta_1=\tau} = \frac{1}{\tau}\mathrm{Cov}_\tau\Big(r(X_\tau), X_\tau\Big) \;\geq\; c.$$

Thus the review menu must provide enough continuation-utility sensitivity not only in total, but also at the margin.

The two conditions in the menu's cutoffs. Because the continuation problem still has four extreme actions, the promised-rent schedule is an increasing step function,

$$r(x) = \sum_{i=1}^{3} \delta_i \mathbf{1}\{x \geq k_i\}, \qquad k_1 < k_2 < k_3,$$

with jumps $\delta_1 = r_D(h)$, $\delta_2 = \widetilde{r}_E(h) - r_D(h)$, and $\delta_3 = M - \widetilde{r}_E(h)$. Let $z_i = (\tau - k_i)/\sqrt{\tau}$. Then the global and marginal first-phase incentive requirements are

$$\underbrace{\sum_i \delta_i \, \Delta(\tau, z_i) \;\geq\; c\tau,}_{\text{global}} \qquad \underbrace{\sum_i \delta_i \, \Lambda(\tau, z_i) \;\geq\; c\tau\,.}_{\text{local}}$$

The first asks whether the menu makes full effort preferable to zero effort. The second asks whether the last increment of first-phase effort is worth taking. This is exactly the same average-versus-marginal distinction that hardened the easy continuation standard.

The ordered menu survives. Attach multipliers $\lambda \geq 0$ and $\mu \geq 0$ to the global and local first-phase constraints. The review-score index becomes

$$Q(x) = \lambda \left[1 - \exp\left(\frac{\tau}{2} - x\right)\right] + \mu \, \frac{x - \tau}{\tau} - 1, \qquad Q'(x) = \lambda \exp\left(\frac{\tau}{2} - x\right) + \frac{\mu}{\tau} > 0. \quad \text{(OA.14)}$$

The important fact is not the extra term itself but its monotonicity. Better review performance still raises the attractiveness of higher-rent actions. Hence the score-by-score problem still selects among the same four actions in the same order. Flexible first-phase effort changes the cutoffs and introduces a second multiplier, but it does not destroy the ordered menu.

One prediction does change. There is one notable implication. In the baseline model the review-score index is bounded above, so the good-stop region appears only for some parameters. If the marginal first-phase constraint binds, then $\mu > 0$ and the linear term in (OA.14) makes $Q(x)$ unbounded above. The good-stop region must then be nonempty.

Economically, marginal first-phase incentives are especially well supplied by very high review outcomes. A low termination threshold can create a large difference between working and not working while doing little to reward the *last* increment of effort. A top prize does the opposite: its covariance with performance is large in the upper tail. Freeing first-phase effort

therefore shifts incentive provision toward the good-stop prize and makes the full four-region ladder generic whenever the local constraint actually bites.

When nothing changes. The extra local constraint need not bind. Since $\Lambda(\tau, z) \geq \Delta(\tau, z)$ exactly when $z \leq z^\dagger(\tau)$, the global condition automatically implies the local one whenever all cutoffs satisfy $k_i \geq k^\dagger(\tau) = \tau - z^\dagger(\tau)\sqrt{\tau} \in (\tau/2, \tau)$. Because the cutoffs are ordered, it is enough to check the lowest one. Thus if the termination threshold is already sufficiently demanding—roughly above half of expected first-phase output—allowing the agent to coast before the review changes nothing.

A step that does not carry over. There is one limitation. For a single continuation threshold, the global and local conditions are jointly sufficient because the agent's loss from deviating is unimodal. Before the review the rent schedule can have three jumps, so the marginal-return function is a Gaussian mixture:

$$\frac{d}{d\eta_1}\mathbb{E}_{\eta_1}\Big[r(X_\tau)\Big] = \frac{1}{\sqrt{\tau}}\sum_i \delta_i\,\phi\left(\frac{k_i - \eta_1}{\sqrt{\tau}}\right).$$

With several peaks, an intermediate effort level can be attractive even when both the global and endpoint-marginal tests pass.

This qualification appears to matter only for long first phases. Numerically, for the optimal menus the mixture is single-peaked on $[0, \tau]$ at every parameter configuration we tried with $\tau \leq 4$, while a second peak appears by $\tau = 9$. Thus for short first phases the two conditions above are exact in the cases examined, whereas late reviews require direct checks of intermediate deviations.

A margin left open. Finally, this section asks whether the principal can still implement full first-phase effort. It does not ask whether she would optimally choose less. Allowing the principal to target an interior first-phase effort would introduce a first-order equality before the review as well and would put the band-contract logic on both sides of the review. We leave that additional margin for future work.

OA.1.6 Proofs

OA.1.6.1 Proof of Lemma OA.1

Proof. Consider minimizing $\pi(0) = \int w f_0$ over $0 \leq w \leq M$ subject to the two equality constraints $\int w f_\eta = \pi_w(\eta)$ and $\int w\,\partial_\eta f_\eta = c$. The feasible set contains w and is the intersection

of the weak-$*$ compact ball with two weak-$*$ closed hyperplanes, hence weak-$*$ compact; the objective is weak-$*$ continuous, so a minimizer $\tilde{w}$ exists. Attaching multipliers λ_1, λ_2 to the two constraints gives a Lagrangian linear in w, which is therefore minimized pointwise over $[0, M]$, with $\tilde{w} = M$ exactly where the index is negative. Dividing that index by $f_\eta(y) > 0$ and using

$$\frac{f_0(y)}{f_\eta(y)} = \exp\left(\frac{\eta^2}{2h} - \frac{\eta y}{h}\right), \qquad \frac{\partial_\eta f_\eta(y)}{f_\eta(y)} = \frac{y-\eta}{h},$$

it becomes

$$\Psi(y) = \exp\left(\frac{\eta^2}{2h} - \frac{\eta y}{h}\right) - \lambda_1 - \frac{\lambda_2}{h}(y-\eta),$$

a strictly convex function of y less an affine one, hence strictly convex. Its negativity set is an interval, which is (OA.4), and since a strictly convex function vanishes at most twice the boundary is null, so no randomization is needed. The first two displayed properties hold because $\tilde{w}$ is feasible for the auxiliary problem and the third because it is optimal. Finally, if w satisfies (G_η) then $\pi_{\tilde{w}}(\eta) - c\eta = \pi_w(\eta) - c\eta \geq \pi_w(0) \geq \pi_{\tilde{w}}(0)$, so $\tilde{w}$ satisfies it too. □

OA.1.6.2 Proof of Lemma OA.2

Proof. That the achievable payments form an interval with the stated endpoints is Lemma OA.1 together with linearity of $\pi(\eta)$ in w on a convex set; the global condition binds at the lower end because lowering $\pi(\eta)$ tightens it. For the upper end let $w^* = M\mathbf{1}\{y \geq \eta - \zeta\sqrt{h}\}$ with $\zeta = \zeta(h)$ and let the agent choose η', writing $v = (\eta' - \eta)/\sqrt{h}$. Then $\pi(\eta') = M\Phi(\zeta + v)$ and, using $c\sqrt{h} = M\kappa$, his payoff is $M\Phi(\zeta + v) - c\eta - M\kappa v$, whose derivative in v is $M[\phi(\zeta + v) - \kappa]$. Since $\phi(\zeta) = \kappa$ and ϕ is symmetric and unimodal, this is positive exactly on $(-2\zeta, 0)$ and negative outside $[-2\zeta, 0]$. The payoff is therefore maximized at $v = 0$ or at the left endpoint $v = -\theta$, so $(\mathrm{IC}_h^{\mathrm{flex}})$ reduces to the single comparison with $\eta' = 0$, namely $D(\theta) := \Phi(\zeta) - \Phi(\zeta - \theta) - \kappa\theta \geq 0$. Now $D(0) = 0$ and $D'(\theta) = \phi(\zeta - \theta) - \kappa$ is positive for $\theta \in (0, 2\zeta)$ and negative for $\theta > 2\zeta$, so $\{\theta \geq 0 : D(\theta) \geq 0\}$ is an interval $[0, \bar{\theta}]$. The hypothesis is exactly $D(\sqrt{h}) \geq 0$, so $\bar{\theta} \geq \sqrt{h}$ and w^* is incentive compatible at every $\eta \leq h$. □

OA.1.6.3 Proof of Lemma OA.3

Proof. Write a deviation as $\eta' = h - u\sqrt{h}$ with $u \in [0, \sqrt{h}]$. The agent's loss from it is $g(u) = M\big[\Phi(z) - \Phi(z - u)\big] - cu\sqrt{h}$, so $(\mathrm{IC}_h^{\mathrm{flex}})$ says $g \geq 0$ on $[0, \sqrt{h}]$. Now $g(0) = 0$ and $g'(u) = M\phi(z - u) - c\sqrt{h}$. As u rises, $z - u$ falls, so $\phi(z - u)$ rises while $z - u > 0$ and falls thereafter: g' is single-peaked. Given $g'(0) \geq 0$, which is $\phi(z) \geq \kappa$, it follows that g' changes sign at most once and only from positive to negative, so g is unimodal and attains its minimum over $[0, \sqrt{h}]$ at an endpoint. Since $g(0) = 0$, $g \geq 0$ if and only if $g(\sqrt{h}) \geq 0$,

which is $M\Delta(h,z) \geq ch$. Conversely, if $\phi(z) < \kappa$ then $g'(0) < 0$ and $g < 0$ just to the right of zero. □

OA.1.6.4 Proof of Lemma OA.4

Proof. By Lemma OA.3 a bar is admissible if and only if $\min\{\Delta(h,z), \Lambda(h,z)\} \geq ch/M$, where $\Lambda(h,z) = \sqrt{h}\,\phi(z)$, so an admissible bar exists exactly when $\overline{\mathcal{M}}(h) \geq ch/M$.

Where the maximum sits. Substituting $s \mapsto z - s$, $\Delta/\Lambda = h^{-1/2}\int_0^{\sqrt{h}} e^{zs-s^2/2}\,ds$, whose derivative in z is $h^{-1/2}\int_0^{\sqrt{h}} s\,e^{zs-s^2/2}\,ds > 0$; the limits are 0 and ∞, so the two cross once, at some $z^\dagger(h)$. At $z = 0$ the ratio is $h^{-1/2}\int_0^{\sqrt{h}} e^{-s^2/2}ds < 1$, so $z^\dagger > 0$. At $z = \sqrt{h}/2$, writing $a = \sqrt{h}/2$, the ratio exceeds one exactly when $2\Phi(a) - 1 > 2a\phi(a)$; the difference vanishes at $a = 0$ and has derivative $2a^2\phi(a) > 0$, so $z^\dagger < \sqrt{h}/2$. Hence $\min\{\Delta, \Lambda\}$ equals Δ, increasing, to the left of $z^\dagger$ and Λ, decreasing on $z > 0$, to its right, and the maximum in (OA.9) is attained at $z^\dagger$. Since $\Delta(h,\cdot)$ is strictly unimodal with peak $\overline{\Delta}(h)$ at $\sqrt{h}/2 \neq z^\dagger$, the inequality $\overline{\mathcal{M}}(h) < \overline{\Delta}(h)$ is strict at every $h > 0$.

The two ends. The admissible bars are $z \in [z_D(h), \zeta(h)]$. For the easy end, $z_E(h) > \sqrt{h}/2 > z^\dagger(h)$ gives $\Lambda(h, z_E) < \Delta(h, z_E) = ch/M = \Lambda\Big(h, \zeta(h)\Big)$, and Λ is decreasing on $z > 0$, so $\zeta(h) < z_E(h)$. For the difficult end, $\Delta(h, z_D) = ch/M \leq \Delta(h, z^\dagger)$ with $\Delta(h,\cdot)$ increasing to the left of $\sqrt{h}/2$ gives $z_D(h) \leq z^\dagger(h)$, whence $\Lambda(h, z_D) \geq \Delta(h, z_D) = ch/M$: the local condition is slack there and $\pi'(h) = M\phi(z_D)/\sqrt{h} > c$. The rent at an admissible bar is $M\Phi(z) - ch$, strictly increasing in z, so the two ends of the segment are the two ends of the interval of bars, which is (OA.10). At $z_D(h)$ the global constraint binds and the rent collapses to $M\Phi(z_D - \sqrt{h})$ by (1); at $\zeta(h)$ it does not, which is why the easy rent must be computed as $\mathbb{E}_h[w] - ch$ rather than as $\mathbb{E}_0[w]$. □

OA.1.6.5 Proof of Lemma OA.5

Proof. Step 1: the jump at the full-effort face. We first show

$$\widehat{P}_{\min}(h) \leq \liminf_{\eta \uparrow h} \widehat{P}_{\min}(\eta), \qquad \text{equivalently} \qquad r_D(h) \leq M\,\mathcal{R}_\kappa(\sqrt{h}).$$

The local condition is an equality below the face and an inequality on it, so the constraint set at $\eta = h$ is the larger one. Formally, take $\eta_n \uparrow h$ and let $w_n \in \widehat{\mathcal{F}}_h(\eta_n)$ attain $\widehat{P}_{\min}(\eta_n)$. The ball $\{0 \leq w \leq M\}$ is weak-$*$ compact, so a subnet converges to some w^*. The maps $\eta \mapsto f_\eta$ and $\eta \mapsto \partial_\eta f_\eta$ are continuous in L^1, so the constraints pass to the limit: $\pi_{w^*}(h) - ch \geq \pi_{w^*}(0)$ and $\pi'_{w^*}(h) = c$, whence $\pi'_{w^*}(h) \geq c$ and $w^* \in \widehat{\mathcal{F}}_h(h)$. Since $\pi_{w^*}(h) = \lim_n \widehat{P}_{\min}(\eta_n)$, the claim follows. Equivalently and more directly: any solution of the pair (OA.5) and (OA.7) at

$\theta = \sqrt{h}$ satisfies (G_η) at $\eta = h$ by construction and (L_η) with equality, hence is admissible at full effort, so its payment bounds $\widehat{P}_{\min}(h)$ from above.

Step 2: a closed form for $\mathcal{R}'_\kappa$. Parametrize the bands satisfying (OA.5) by a, with $b = b(a)$ solving $\phi(b) = \phi(a) - \kappa$, and write $E_x = \phi(x+\theta)/\phi(x) = e^{-\theta x - \theta^2/2}$. Differentiating $\phi(b) = \phi(a) - \kappa$ and using $\phi'(x) = -x\phi(x)$ gives $b'(a) = a\phi(a)/\Big(b\phi(b)\Big)$. Write $P(a) = \Phi(b) - \Phi(a)$, $\mathcal{R}(a,\theta) = \Phi(b+\theta) - \Phi(a+\theta)$ and $G(a,\theta) = P(a) - \mathcal{R}(a,\theta) - \kappa\theta$ for the slack in (G_η). Then

$$P'(a) = \phi(b)b'(a) - \phi(a) = \phi(a)\,\frac{a-b}{b} \;<\; 0,$$

so minimizing the payment means taking the largest admissible a; call it $a^*(\theta)$, at which (G_η) binds. Substituting $\phi(x+\theta) = \phi(x)E_x$,

$$\mathcal{R}_a = \phi(a)\left(\frac{a}{b}E_b - E_a\right), \qquad \mathcal{R}_\theta = \phi(b)E_b - \phi(a)E_a,$$
$$G_a = \frac{\phi(a)}{b}\Big[a(1-E_b) - b(1-E_a)\Big].$$

Differentiating $G(a^*(\theta),\theta) = 0$ and $\mathcal{R}_\kappa(\theta) = \mathcal{R}(a^*(\theta),\theta)$, and using $G_\theta = -\mathcal{R}_\theta - \kappa$ and $G_a = P' - \mathcal{R}_a$,

$$\mathcal{R}'_\kappa = \mathcal{R}_\theta - \mathcal{R}_a\frac{G_\theta}{G_a} = \frac{P'\mathcal{R}_\theta + \kappa\,\mathcal{R}_a}{P' - \mathcal{R}_a}.$$

Expanding the numerator and substituting $\kappa = \phi(a) - \phi(b)$, the terms in E_a and E_b collect as $P'\mathcal{R}_\theta + \kappa\mathcal{R}_a = \frac{\phi(a)}{b}\Big(a\phi(a) - b\phi(b)\Big)(E_b - E_a)$, so dividing by G_a,

$$\mathcal{R}'_\kappa(\theta) = \frac{\Big(a\phi(a) - b\phi(b)\Big)\Big(E_b - E_a\Big)}{a\,(1-E_b) - b\,(1-E_a)}. \tag{OA.15}$$

Step 3: signing it. Since $\phi(b) < \phi(a)$ we have $|b| > |a|$, which together with $b > a$ forces $b > 0$; as $\theta > 0$, $E_b < E_a$ and the second factor of the numerator is negative. Differentiating $\pi(\eta') = M[\Phi(b+\theta') - \Phi(a+\theta')]$ twice at $\theta' = 0$ gives $\pi''(\eta) = \frac{M}{h}\Big(a\phi(a) - b\phi(b)\Big)$, so the first factor is nonpositive precisely when the minimizing band satisfies (S_η); Remark OA.1 verifies that it does, so this band lies in $\widehat{\mathcal{F}}_h(\eta)$ and attains $\widehat{P}_{\min}(\eta)$. Finally, the denominator equals $b\,G_a/\phi(a)$ with $\phi(a) > 0$ and $b > 0$; since a^* is the largest a at which $G \geq 0$, and $G(a^*,\theta) = 0$, we have $G_a(a^*,\theta) \leq 0$. The numerator of (OA.15) is therefore nonnegative and the denominator nonpositive, so $\mathcal{R}'_\kappa \leq 0$.

Conclusion. Since $\mathcal{R}_\kappa \geq 0$ and $\mathcal{R}'_\kappa \leq 0$,

$$\frac{d}{d\theta}\left[\frac{\mathcal{R}_\kappa(\theta)}{\theta}\right] = \frac{\theta\,\mathcal{R}'_\kappa(\theta) - \mathcal{R}_\kappa(\theta)}{\theta^2} \;\leq\; 0.$$

Hence for $\theta = \eta/\sqrt{h} < \sqrt{h}$, using (OA.13) and then Step 1,

$$\frac{\rho(\eta)}{\eta} = \frac{M\,\mathcal{R}_\kappa(\theta)}{\theta\sqrt{h}} \;\geq\; \frac{M\,\mathcal{R}_\kappa(\sqrt{h})}{h} \;\geq\; \frac{r_D(h)}{h} = \frac{\rho(h)}{h},$$

which is (OA.12). □

Remark OA.1. The two signs invoked in Step 3 of the proof are economic rather than technical. That $a\phi(a) \leq b\phi(b)$ is the agent's second-order condition: at the difficult end of the band family the contract would make the target a local *minimum* of the agent's payoff, and such contracts implement nothing. That $a(1-E_b) \leq b(1-E_a)$ says the payment-minimizing band sits on the boundary of the region where the agent still prefers the target to abandoning effort; pushing the bar any further would break that constraint. The two never conflict: at the extreme $a = \zeta(h)$ the band degenerates to an upper-tail bonus, where $a\phi(a) - b\phi(b) = \zeta(h)\kappa > 0$ and (S_η) fails outright, so the binding restriction on a is always (G_η). Numerically, over the entire admissible range $0 < \kappa < \phi\big(z^\dagger(\theta^2)\big)$ and θ up to 12, the global constraint binds at every optimum and (S_η) is strictly slack.

OA.1.6.6 Proof of Proposition OA.1

Proof. Let $\widehat{\mathcal{A}}(h)$ be defined as in (OA.3) with $[\widehat{P}_{\min}, \widehat{P}_{\max}]$ in place of $[P_{\min}, P_{\max}]$. By (OA.1), $\mathcal{A}^{\text{flex}}(h) \subseteq \widehat{\mathcal{A}}(h)$.

We first show $\widehat{\mathcal{A}}(h)$ lies in the quadrilateral with vertices A_B, A_D, $\widetilde{A}_E$, A_G. Because $(\eta, P) \mapsto (r, s)$ is linear and invertible, it is enough to show the feasible (η, P) region lies in the quadrilateral with vertices $(0,0)$, $(h, \widehat{P}_{\min}(h))$, $(h, \widehat{P}_{\max}(h))$ and $(0, M)$. The region lies between the graphs of $\widehat{P}_{\min}$ and $\widehat{P}_{\max}$, so it suffices to bound each.

Upper edge. By (OA.6), $\widehat{P}_{\max}(\eta) \leq M\Phi(\zeta(h)) = \widehat{P}_{\max}(h)$, while the upper edge of the quadrilateral is the chord from $(0, M)$ to $(h, \widehat{P}_{\max}(h))$, which lies weakly above that constant because $M \geq \widehat{P}_{\max}(h)$. This step is unconditional.

Lower edge. The lower edge is the chord from $(0,0)$ to $(h, \widehat{P}_{\min}(h))$, so the requirement is $\widehat{P}_{\min}(\eta) \geq (\eta/h)\widehat{P}_{\min}(h)$, which is Lemma OA.5. By Lemma OA.4, $\widehat{P}_{\min}(h) = M\Phi(z_D(h))$ and $\widehat{P}_{\max}(h) = M\Phi(\zeta(h))$, so the quadrilateral is exactly $\text{co}\{A_B, A_D, \widetilde{A}_E, A_G\}$.

It remains to place the four vertices in $\mathcal{A}^{\text{flex}}(h)$. A_B and A_G are the constant contracts $w \equiv 0$ and $w \equiv M$, covered by the no-effort regime of Section OA.1.2; A_D and $\widetilde{A}_E$ are covered by Lemma OA.4(ii). Hence

$$\text{co}\big\{A_B, A_D, \widetilde{A}_E, A_G\big\} \subseteq \text{co}\,\mathcal{A}^{\text{flex}}(h) \subseteq \text{co}\,\widehat{\mathcal{A}}(h) = \text{co}\big\{A_B, A_D, \widetilde{A}_E, A_G\big\},$$

and the chain closes. □

Remark OA.2. Because (OA.13) eliminates the horizon, the whole argument can be checked exhaustively rather than sampled. Over the entire admissible range $0 < \kappa < \phi\big(z^\dagger(\theta^2)\big)$ — the condition $\overline{\mathcal{M}}(h) > ch/M$ with $h \geq \theta^2$, by (OA.9) and monotonicity of $z^\dagger$ — and θ up to 12, which covers every (h, c, M) satisfying $\overline{\mathcal{M}}(h) > ch/M$ with $h \leq 144$: the closed form (OA.15) agrees with numerical differentiation to six decimals, both sign conditions of Step 3 hold at every point, and the jump of Step 1 is strictly positive throughout.

OA.2 Continuous observability

This appendix supplies the recursive formulation and verification behind Proposition 8, together with the numerical comparison reported in Section 6.2. The primitives are unchanged except that the entire path of output is public and effort is chosen pointwise. Thus

$$dX_t = e_t\,dt + dB_t, \qquad e_t \in \{0, 1\},$$

and both the recommended effort process and the terminal payment $Y \in [0, M]$ may depend on the public output history.

OA.2.1 Promised utility and incentive compatibility

Let

$$W_t = \mathbb{E}_t\left[Y - c\int_t^T e_s\,ds\right] \tag{OA.16}$$

be the agent's continuation utility under the recommendation. Martingale representation gives an adapted process β_t such that

$$dW_t = ce_t\,dt + \beta_t(dX_t - e_t\,dt), \qquad W_T = Y. \tag{OA.17}$$

If the agent deviates from $e_t = 1$ to $e_t = 0$ over an infinitesimal interval, he saves flow cost c and changes the drift of promised utility by $-\beta_t$. Hence work is locally optimal if and only if $\beta_t \geq c$; similarly, shirking is locally optimal only if $\beta_t \leq c$. This is the familiar continuous-time incentive condition in Sannikov (2008).

The principal's expected payoff can be written entirely in terms of expected work and the agent's initial promise. Taking expectations in (OA.16) at date zero gives $\mathbb{E}Y = W_0 + c\mathbb{E}\int_0^T e_t dt$, and therefore

$$\Pi = (1 - c)\mathbb{E}\int_0^T e_t\,dt - W_0. \tag{OA.18}$$

For a fixed W_0, the principal therefore wants to maximize expected time spent working.

The payment bounds imply absorbing boundaries for promised utility.

Lemma OA.6. *In any incentive-compatible contract, $W_t \in [0, M]$ at every history. If $W_t = 0$, no further effort can be induced and the continuation payment is zero. If $W_t = M$, no further effort can be induced and the continuation payment is M.*

Proof. From date t, shirking forever costs the agent nothing and yields a nonnegative payment, so incentive compatibility implies $W_t \geq 0$. Also $W_t \leq \mathbb{E}_t Y \leq M$. If $W_t = 0$, the deviation to permanent shirking must yield zero, which forces $Y = 0$ almost surely. Then (OA.16) implies zero remaining effort. If $W_t = M$, then $M = \mathbb{E}_t Y - c\mathbb{E}_t \int_t^T e_s ds \leq M - c\mathbb{E}_t \int_t^T e_s ds$, again forcing zero remaining effort, and $Y = M$ almost surely. □

OA.2.2 The recursive problem and the band contract

Scale promised utility by effort cost: $u = W/c$ and $m = M/c$. Let $v(h, u)$ be the maximal expected amount of work that can still be induced when h units of time remain and the scaled promise is $u \in [0, m]$. The recursive equation is

$$v_h = \max \left\{ \sup_{b \geq 1} \left[1 + v_u + \frac{1}{2} b^2 v_{uu} \right], \sup_{b \leq 1} \frac{1}{2} b^2 v_{uu} \right\}, \tag{OA.19}$$

with $v(h, 0) = v(h, m) = 0$ and $v(0, u) = 0$. Here $b = \beta/c$. The first term recommends work; the second recommends shirking.

Consider the candidate that always works in the interior with the smallest incentive-compatible sensitivity, $b = 1$. Under this policy the scaled promise follows

$$dU_t = dt + dB_t$$

until it exits $(0, m)$. If $\sigma = \inf\{t : U_t \notin (0, m)\}$, its expected work time is

$$v(h, u) = \mathbb{E}_u[\sigma \wedge h],$$

which solves

$$v_h = 1 + v_u + \frac{1}{2} v_{uu}, \qquad v(h, 0) = v(h, m) = v(0, u) = 0. \tag{OA.20}$$

Proposition OA.2. *The solution to (OA.20) satisfies $v_h \geq 0$ and $v_{uu} \leq 0$. Consequently the interior work policy with $b = 1$ solves the HJB (OA.19). Hence the optimal contract works until promised utility hits either 0 or M and uses $\beta = c$ while it remains in the interior.*

Proof. Differentiate (OA.20) with respect to h. The derivative $k = v_h$ solves the same homogeneous parabolic equation in the interior, with $k = 0$ at the spatial boundaries and initial value $k(0, u) = 1$. The maximum principle gives $k \geq 0$.

For concavity, let $g = v_{uu}$. Again g solves the homogeneous parabolic equation. Its initial value is zero. At the lower boundary, $v(h, 0) = 0$ implies $v_h(h, 0) = 0$, and (OA.20) yields $g(h, 0) = -2[1 + v_u(h, 0)] \leq 0$ because $v \geq 0$ and hence $v_u(h, 0) \geq 0$. At the upper boundary, a unit-drift Brownian motion starting ε below m reaches m in expected time ε if the lower boundary is ignored, so $v(h, m - \varepsilon) \leq \varepsilon$ and $v_u(h, m) \geq -1$. Therefore $g(h, m) = -2[1 + v_u(h, m)] \leq 0$. The maximum principle gives $g \leq 0$ everywhere.

With $v_{uu} \leq 0$, the work term in (OA.19) is maximized at the smallest admissible b, namely $b = 1$, and equals v_h by (OA.20). The shirking term is at most zero, while $v_h \geq 0$. Thus the candidate solves the HJB. Standard verification then implies optimality. □

Returning to dollars, promised utility satisfies

$$W_t = W_0 + cX_t$$

until the first time it hits 0 or M. The lower hit produces firing with zero payment and the upper hit produces a buyout at M. Conditional on no hit by T, the agent is paid W_T. By (OA.18), the initial promise solves

$$u^* \in \arg\max_{u \in [0,m]} \{(1 - c)v(T, u) - cu\}. \tag{OA.21}$$

Concavity of v makes the objective concave. An interior optimum satisfies

$$(1 - c)v_u(T, u^*) = c. \tag{OA.22}$$

For numerical work, v can be computed directly from (OA.20). A useful closed form follows from the killed heat kernel. With $k_n = n\pi/m$ and $\lambda_n = (1 + k_n^2)/2$,

$$v(T, u) = \frac{4}{m} e^{-u} \sum_{n \geq 1} \sin(k_n u) \frac{k_n[1 - (-1)^n e^m][1 - e^{-\lambda_n T}]}{(1 + k_n^2)^2}. \tag{OA.23}$$

The PDE representation is numerically more stable for large m, where cancellation in the series becomes important.

OA.2.3 Comparison with one review

The continuous-observation environment is not a superset of the baseline environment: the agent now observes the same running performance information as the principal and can adjust effort accordingly. The comparison therefore measures the joint effect of finer monitoring and finer agent responsiveness.

For the benchmark $T = 1.5$, $c = 0.6$, and $M = 2$, direct evaluation gives

	No review	One optimal review	Continuous observation
Principal payoff	0.1949	0.2389	0.1345
Agent rent	0.405	0.268	0.207
Expected work time	1.500	1.268	0.854

The continuous contract gives the agent the smallest rent, but obtains much less production because the promised-value process hits an absorbing boundary with substantial probability. This is why one review can dominate continuous monitoring even though the latter provides finer information.

Varying M at $T = 1.5$ and $c = 0.6$ produces three useful thresholds. The band contract becomes profitable at approximately $M = 0.670$. The best contract with at most one review becomes profitable at approximately $M = 1.373$, and its value crosses the band value at approximately $M = 1.488$. Finally, the no-review full-effort contract becomes implementable at approximately $M = 1.958$. Thus the band contract is most valuable under very tight caps; for moderate caps the one-review contract dominates it, and for sufficiently loose caps the optimal one-review contract converges to no review.

OA.2.4 Why the lower boundary cannot be reflected

The lower absorbing boundary distinguishes hidden effort from the dynamic-participation problem in McClellan (2022). There, a moving approval standard can keep an experimenting agent exactly at his participation constraint while experimentation continues. Here that construction conflicts with hidden-action incentives.

Proposition OA.3. *Let $\sigma = \inf\{t : W_t = 0\}$. Under any incentive-compatible contract, effort must be zero after σ almost surely. In particular, promised utility cannot be reflected at zero while positive effort continues.*

Proof. Lemma OA.6 already gives the result. The local intuition is useful. If work continued just after σ, then (OA.17) and incentive compatibility would imply, over a short interval,

$$W_{\sigma+s} = cs + \int_{\sigma}^{\sigma+s} \beta_t \, dB_t, \qquad \beta_t \geq c.$$

The martingale term has quadratic variation at least $c^2 s$ and therefore takes negative values of order $\sqrt{s}$ arbitrarily close to zero. The deterministic drift is only order s, so with positive probability W immediately becomes negative, contradicting limited liability. Thus zero must be absorbing whenever hidden effort is being induced. □

In McClellan (2022), by contrast, the agent controls participation rather than hidden effort. At a binding participation boundary, continuation value can touch zero with zero sensitivity to new information. Our incentive constraint rules out that smooth-pasting response because the sensitivity of promised utility must remain at least c whenever the agent works. This distinction explains why the continuous-observation contract terminates rather than relaxing the performance standard after a bad enough history.

OA.3 Random review timing with pointwise effort

This appendix supplies the proof and numerical details for the two-date pointwise-effort experiment in Section 6.3. Continuation actions after review are the pointwise-feasible actions of Section 6.1, so $\widetilde{A}_E$ replaces A_E.

OA.3.1 Two dates under pointwise effort

Proof of Proposition 9. Dynamic reduction. Before τ_1 no review can occur, so the agent receives no information about the realized branch, and because the review-score law and the linear effort cost depend on the pre-τ_1 path only through total effort, any deviation is summarized by $e \in [0, \tau_1]$. If the early review occurs his utility is $U_1(e)$. If it does not, he learns the date is τ_2; starting from e, feasibility confines cumulative effort at τ_2 to $[e, e+d]$, so his continuation value is $\widehat{U}_2(e)$ in (11). Choosing e is therefore worth $pU_1(e) + (1-p)\widehat{U}_2(e)$. On path $e = \tau_1$, and obedience after non-arrival requires cumulative effort τ_2, which is $U_2(\tau_2) = \widehat{U}_2(\tau_1)$. Given that, full effort before τ_1 is optimal if and only if $pU_1(\tau_1) + (1-p)U_2(\tau_2) \geq pU_1(e) + (1-p)\widehat{U}_2(e)$ for every e, which rearranges to (12).

The bound. The late branch induces full effort on its own when $U_2(\tau_2) \geq U_2(m)$ for all $m \in [0, \tau_2]$; then $\widehat{U}_2(e) \leq U_2(\tau_2)$, so $D_2(e) \geq 0$; the same condition applied at $m = \tau_1$ gives $U_2(\tau_2) = \widehat{U}_2(\tau_1)$. Where $D_1(e) \geq 0$, (12) holds for every p. Where $D_1(e) < 0$ it is equivalent to $p[D_2(e) - D_1(e)] \leq D_2(e)$, that is $p \leq D_2(e)/[D_2(e) - D_1(e)]$; deterring all deviations is therefore $p \leq \overline{p}$ in (13).

Dominance. Under full effort the lottery is worth $V(p) = P_2 + p(P_1 - P_2)$, affine and increasing in p, so $V(p) > V^{D,\mathrm{pw}}$ if and only if $p > \underline{p}$. A feasible strict improvement exists exactly when $(\underline{p}, 1]$ meets $[0, \overline{p}]$, which is (14). □

OA.3.2 Numerical construction

All figures are direct evaluations of the Gaussian formulas; no simulation is used. Set $T = 1.5$, $c = 0.72$, $M = 2.60$ throughout. Numbers are rounded to four decimals (five for the quantities also reported in Section 6.3); the replication files report them in full precision.

Pointwise example. Take $\tau_1 = 0.0949$ and $\tau_2 = 0.8166$, so $h_1 = 1.4051$ and $h_2 = 0.6834$. With $z_D(h)$ the difficult root of $\Delta(h, z) = ch/M$ and $\zeta(h)$ the positive root of $\phi(\zeta) = c\sqrt{h}/M$ from (9),

$$r_D(h) = M\Phi\Big(z_D(h) - \sqrt{h}\Big), \qquad \widetilde{r}_E(h) = M\Phi\Big(\zeta(h)\Big) - ch,$$

giving $z_D = 0.0358$, $\zeta = 0.6245$, $r_D = 0.3254$, $\widetilde{r}_E = 0.8963$ on the early branch and $z_D = -0.6432$, $\zeta = 1.0540$, $r_D = 0.1841$, $\widetilde{r}_E = 1.7285$ on the late branch. The menus use cutoffs

	τ_i	k_{i1}	k_{i2}	k_{i3}
early	0.0949	−0.0450	0.8789	0.9705
late	0.8166	0.4048	1.1470	1.5035

so that the rent schedule is the step function with jumps $\delta_{i1} = r_D(h_i)$, $\delta_{i2} = \widetilde{r}_E(h_i) - r_D(h_i)$ and $\delta_{i3} = M - \widetilde{r}_E(h_i)$, and

$$U_i(e) = \sum_{j=1}^{3} \delta_{ij}\Phi\left(\frac{e - k_{ij}}{\sqrt{\tau_i}}\right) - ce, \qquad U_i'(e) = \frac{1}{\sqrt{\tau_i}}\sum_{j=1}^{3} \delta_{ij}\phi\left(\frac{e - k_{ij}}{\sqrt{\tau_i}}\right) - c. \tag{OA.24}$$

Under full effort the region probabilities are $(0.3249, 0.6696, 0.0032, 0.0022)$ on the early branch and $(0.3243, 0.3183, 0.1338, 0.2236)$ on the late branch, whence $P_1 = 0.13301$ and $P_2 = 0.03199$.

For the sequential check, $U_1(\tau_1) = 0.1583$ and $U_1(0) = 0.1842$, so $D_1(0) = -0.0259$, and $U_1' \in [-0.2750, -0.2721]$ on $[0, \tau_1]$: the early branch by itself strictly favors less pre-review effort throughout. On the late branch $U_2(\tau_2) = 0.2832$ while U_2' has a unique zero on $[0, \tau_2]$ at $m_0 = 0.3592$, a strict minimum with $U_2(m_0) = 0.2152$; hence $U_2(\tau_2)$ is the global maximum and the late branch is self-incentive-compatible. Because U_2 has only this interior minimum, $\widehat{U}_2(e) = \max\{U_2(e), U_2(e + d)\}$, with the two terms crossing once at $e_c = 0.0021$. At $e = 0$ the agent prefers not to add effort after learning the date, so $\widehat{U}_2(0) = U_2(0) = 0.2599$ and $D_2(0) = 0.0233$. The deviation-specific bound $q(e) = D_2(e)/[D_2(e) - D_1(e)]$ is increasing on $[0, e_c]$ and on $[e_c, \tau_1)$, so the binding deviation is $e = 0$ and

$$\overline{p} = q(0) = \frac{D_2(0)}{D_2(0) - D_1(0)} = 0.47293.$$

A relaxation of the deterministic pointwise problem has value 0.07850, so $V^{D,\mathrm{pw}} \leq 0.07850$ and $\underline{p} \leq 0.46047 < \overline{p}$. At $p = 0.465$ the worst deviation still loses $0.465D_1(0) + 0.535D_2(0) = 0.00039 > 0$, and the contract is worth $0.465P_1 + 0.535P_2 = 0.07896$, exceeding the deterministic upper bound by 0.00046.